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Detailed Project Document

GREEN ECONOMIC DEVELOPMENT PROJECT

November, 2014.

Short document description

The Green Economic Development detailed project proposal document is developed in a manner to firstly provide the reader with key SWEDEN's project contribution effects (I SUMMARY), followed by a detailed description of current policy and legislative processes, as well as challenges facing the energy efficiency environment in Bosnia and Herzegovina, with focus on public sector buildings (II ENERGY EFFICIENCY IN THE PUBLIC SECTOR BUILDINGS IN BOSNIA AND HERZEGOVINA – POLICY AND CHALLENGES). Chapter III (GREEN ECONOMIC DEVELOPMENT PROJECT) provides detailed overview and introduces the reader to GED project goals, components, activities and the inter-linkage of activities. Chapter IV (SWEDEN's INVOLVEMENT) identifies project components which would significantly contribute from SWEDEN's involvement, and provides detailed overview of envisaged budget, sustainability and effects on the project. Chapter V (PROJECT ARRANGEMENTS – MANAGEMENT & REPORTING) provides an overview of Management and Reporting arrangements. Logical framework and Results matrix are provided in Annexes.

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I. SUMMARY - SWEDEN'S PROJECT CONTRIBUTION RESULTS

SWEDEN's involvement in the Green Economic Development project (3.6m € within three years) would result in:

- motivating end-users to enter into EE investments (and later into the newly established financial mechanisms) while further contributing to leverage and unlock additional funds from end-users
- new financial mechanism established within Fund and proof provided to BiH based financial sector (government and commercial) that energy efficiency investments are economically and financially attractive and cost-effective
- at least **10 additional detailed energy audits conducted per year**
- **15 additional infrastructure projects implemented per year**
- about **195.000 € additional annual energy costs savings** achieved
- higher co-financing amount from end-users (matching 1:1, or leveraging up to two times) – **about 846,000 € co-financing by end-users/project partners** ensured per year
- **additional 260 man-months direct employment/"green jobs" generated per year**
- **additional 865 tons of CO_2 emission reduced annually**
- improved execution, coverage and outreach of GED's marketing campaign
- scaling-up and creation self-sustainability within energy efficiency related activities in BiH and contribution to BiH's EU accession and fulfilment of Energy Community Treaty obligations

Short-term aim:

- create interest and motivate public sector buildings' end-users to invest into energy efficiency by promoting grant co-financing
- develop capacity and skills of Environmental Funds (systematic approach to decision making processes of energy efficiency investments)
- achieve budget cost savings through implementation of EE projects in public sector buildings and reinvest savings into energy efficiency or other infrastructure projects

Mid-term aim:

- institutionalize energy monitoring and reporting mechanisms / energy management
- generate employment of domestic workforce
- increase public awareness and understanding of EE benefits
- develop and adopt sustainable financial mechanisms within Environmental Protection Funds for EE project financing (performance based granting, revolving mechanisms, soft-loans, etc.)
- contribute to BiH's EU accession (EPBD and EED directives)
- lower the risk to enter EE investments/provide loans for EE investments in BiH

II. ENERGY EFFICIENCY IN THE PUBLIC SECTOR BUILDINGS IN BOSNIA AND HERZEGOVINA – POLICY AND CHALLENGES

2.1. INTRODUCTION

From an energy consumption perspective, Bosnia and Herzegovina is characterized as a country with very high inefficiency within the residential, non-residential/public, industry and service sector. At the same time however, Bosnia and Herzegovina has one of the most significant energy conservation potentials in the region and could base its further mid-term economic development and generation of new employment on energy efficiency improvement measures in the residential and public sector. Currently, BiH consumes about 20% of its GDP on energy[1], which is three times higher than in the U.S. and EU countries. According to the Report "Regular Review of Energy Efficiency Strategies in BiH" prepared under the obligations of the Energy Charter Treaty, gross total primary energy consumed per unit of GDP is 0.938 toe / USD 2000[2], which is 2.5 times the average of 27 EU countries and higher than almost any other country in the SEE region. Moreover, the average energy consumption of a public building in Bosnia and Herzegovina is three times higher than the EU average, categorizing them as *completely energy inefficient buildings* (in accordance to EU Eco-Management and Audit Scheme – EMAS). In order to meet this energy intensive consumption demand, a significant amount of budget funds must be allocated for energy expenditures of public buildings (educational, health, cultural, municipal and entity/state institutions etc.) representing a major proportion of the already inadequate public budget.

Most of the western economies base their economic development on strategic documents incorporating energy efficiency measures in the public and residential sector and the utilization of renewable energy sources, while contributing to fight climate change, fulfil EU and UNFCCC multilateral binding agreements and goals, and simultaneously generate new employment through mid/large scale public works. Unemployment is a challenge that has long been present in BiH as a consequence of war devastation and transition to a market economy. In recent years it has been exacerbated by the impact of the global economic crisis on the country's economy (currently official unemployment rate of 28%[3], according to ILO methodology). While some efforts are being made to increase employment by encouraging investment in strategic projects such as energy facilities and systematic decrease of energy consumption, the potential of these investments which have been traditionally linked with the concept of public works employing a sizeable proportion of workforce is currently significantly limited, primarily due to a number of non-existing legislative commitment factors within the energy efficiency / renewable energy primary and secondary legislation and undeveloped MVR (monitoring-verification-reporting), market and financial mechanisms.

Although the "Support to BiH to meet the requirements of the Energy Community Treaty with special focus on EE and RE" project, financed by the European Union, had the goal to support BiH in meeting the requirements of the Energy Community Treaty, within the European Commission's 2012 published "Bosnia And Herzegovina 2012 Progress Report" it has been stated that "there was little

[1] World Bank data
[2] Regular Review of Energy Efficiency Strategies in BiH, 2011
[3] BiH Agency for Statistics

progress on energy efficiency" [4]! Namely, the legislation on EE / RES has partially been drafted and partially adopted. Entity's and/or State level energy efficiency action plan(s) and a credible roadmap(s) for transposition of the relevant EU legislation on energy efficiency to meet obligations under the Energy Community Treaty are still missing. Moreover, "the systematic diffusion of energy efficiency considerations as an integral part of policy, regulation and control has hardly begun in BiH" [5], preventing the creation of a favourable investing environment and systematic energy monitoring and decision making approach into EE / RES investments.

The energy sector in BiH is organized in accordance to the General Framework Agreement for Peace in Bosnia and Herzegovina, positioning entity line ministries as the key players in the EE / RES framework while giving the state-wide level, represented by the Ministry of Foreign Trade and Economic Relations (MOFTER), a coordinating/reporting role for multilateral binding agreements. As such, the direct implementation of EE / RES related activities and EU aquies (such as achieving energy saving targets, energy monitoring, enforcements of legislations, financing mechanisms etc.) is to be carried out on the entity/cantonal levels. At this time Bosnia and Herzegovina has no official Energy Policy or Strategy and is not in the position to purposefully absorb or allocate required funds for EE investments. As of now, development and capacity building projects to meet BiH's EE / RES EU accession obligations focused solely on the state level while the involvement of entity and cantonal level authorities has been limited, resulting in limited success of EE/RES projects.

Due to stated facts, due to limited and therefore relatively slow state-level planning processes, the complexity of state-entity interactions and the country's administrative divisions and responsibilities, this situation analysis argues that further enforcement of energy efficiency / renewable energy in BiH should focus on a bottom-up (local-ministries/canton/entity-state) rather than top-down (state-ministries/canton/entity-local) approach while creating MVR, market and financial EE / RES mechanisms in Bosnia and Herzegovina. This would contribute to faster the creation of an environment attractive for EE /RES investments in the public and residential sector, generation of new employment and would result in the creation of clear energy monitoring and target achieving mechanisms within the country.

[4] Bosnia And Herzegovina 2012 Progress Report, EC, 2012
[5] In-Depth Review ofEnergy Efficiency Policies and Programmes, ECT, 2012

2.2. CURRENT SITUATION AND MAJOR CHALLENGES

2.2.1. EE / RES in public and residential buildings

In accordance to conducted energy studies, most of the energy in Bosnia and Herzegovina is consumed within the public and residential building sector – about 55% of total final energy consumption. The rest is consumed by the industry, service and transportation sector. The energy characteristic of public and residential sector is described as very energy-intensive due to high energy inefficiency. The average public building in BiH is consuming 220 kWh/m^2 for thermal heating purposes, while the average residential building consumes 180 kWh/m^2 [6]. Both are categorizing them as *completely energy inefficient buildings* with strong energy conservation potential. Moreover, about 90%[7] of the current BiH building stock does not meet the current technical standards in BiH. Due to its inefficiency, the average public building in Bosnia and Herzegovina can achieve an energy consumption reduction of up to 60% with investments resulting in a rather short pay-back period (up to six years). However, a systematic approach and allocation of investments into energy efficiency and the utilization of renewables is missing.

There are several factors influencing the present energy (in)efficiency situation in BiH – however, the two most damping forces are the current energy prices (primarily for residential buildings) and the lack of binding legislative commitment factors within the energy efficiency / renewable energy primary and secondary legislation (primarily for public buildings). Namely, the energy pricing policy in BiH is not market driven but politically influenced and reflected though social-cross subsidizing resulting in one of the lowest energy price markets in the region and Europe. There are no emission taxes for fossil fuels, electricity generated to large extent from fossil fuels is often used for heating purposes and district heating services are usually paid on a fix-term basis while consumption based billing is very rare. These facts do not contribute to a motivating environment to invest in energy efficiency and renewable energy sources in the residential sector by the general public. On the other hand, the public sector should lead by example the *increasing of energy efficiency and utilization of renewable energy* processes in the country and communicate its benefits to the wider audience. However, in the absence of clear legislation framework, strategy and action plans, as well as reporting and monitoring mechanisms, which all together shall enable an investing environment, a systematic and comprehensive undertaking to increase energy efficiency and utilize renewables in the public sector in BiH will not occur! Moreover, one major concern is that, once adopted, the primary and secondary legislation framework drafted for FBiH and RS will not be implementable due to lack of human and technical resources, lack of understanding and planning on various authority/government levels, as well as fragmented energy related cost covering jurisdictions (and therefore responsibilities) of public buildings throughout BiH. Since the legislation on EE / RES is only partially drafted and has not been fully adjusted to BiH context during the drafting stage, once adopted (Law on EE adopted in RS, in FBiH to be adopted by the end of 2014), an additional concern is that it will not be ready for implementation and will require additional interventions by line Ministries. The most challenging and basic issue to address energy efficiency and utilization of renewable energy in BiH public buildings is however the non-existing data and overview of current

[6] BiH Energy Efficiency GAP analysis, S. Avdić, 2012
[7] Energy sector study in BIH, 2008

public building stock. Namely, due to the fragmented and complex inter-authority jurisdictions, especially in FBiH, authorities and line ministries do not possess a clear overview (in most cases even amount) of public buildings under their jurisdiction, not to mention energy and water related consumption and the costs they bear on a monthly basis.

Due to omission of lower level governments during and after EE / RES legislation drafting, due to the fragmented jurisdiction of different types of public buildings and not present communication and knowledge and information sharing on expected obligations and actions which will be required to be undertaken, it is highly recommended that further EE / RES activities focus on a bottom-up approach in order to be in the position to achieve and communicate energy savings while creating sustainable energy monitoring and reporting mechanisms on entity and state level in BiH and an systematic approach to decision making process on energy efficiency and renewables. This would highly contribute to BiH obligations under the Energy Community Treaty.

2.2.2. Monitoring and reporting on energy consumption and emissions in public sector buildings

One of the main challenges various levels of governments are facing throughout BiH is the identification and allocation mechanisms of energy and water related costs of public buildings. Namely, the vast number of governments/ministries with jurisdiction of energy and water costs payment does not have an overview of the public building type, the amount of energy and water related costs nor are the expenditures monitored, recorded or analyzed.

In accordance to the expected energy related goals/target communicating (top-down) and implementation achieving (bottom-up) mechanisms, which shall be backed-up by EE / RES legislation adoption, public buildings under ministerial, cantonal and municipal jurisdiction shall increase their energy efficiency and utilize renewable energy in accordance to yet to-be-developed actions plans. However, the concern is that the present situation and understanding of BiH's ECT obligations on authority levels below the entity level and outside EE / RES legislation related line ministries (the monitoring and reporting on energy consumption, energy savings, emissions etc.) is non-existing or very limited, and is by far not able to absorb and implement the required activities. Furthermore, without an existing baseline overview on current energy consumption and emissions, in the absence of state level statistical data and reliable and relevant data from entities, a clear and well structured energy monitoring and reporting mechanisms cannot be implemented and BiH's obligations to multilateral agreements only partially, if at all, met. The non-existence of statistical data favors a bottom-up approach over unreliable top-down estimations.

2.2.3. Utility prices & Financing EE / RES activities

Bosnia and Herzegovina has a relatively low living standard (GDP per capita of 70 per cent below the EU-27 average[8], 23,9% of rural households and 11% of urban households are poor[9], it is 101st in

[8] Eurostat
[9] National Human Development Report 2013, UNDP, 2013

world rankings according to Inflation/Average Consumer Prices[10]), with *average income per households* of net USD 558 and a *consumption basket* of USD 913. The average distribution of living expenses on utilities in BiH amounts 10% of total expenses. These factors, among others, influence that utility prices (energy and water) are kept artificial low by authorities.

The majority of district heating and water utility companies in BiH are public companies funded by municipal or cantonal authorities. Although those companies in BiH are legal bodies with full responsibilities and obligations in accordance to the local and entity commercial law, and other binding legislations, they have only limited influence to the pricing policy for their services. Namely, since companies are in full ownership of local/cantonal authorities, all prices for the different tariff groups (households, commercial buildings, public buildings) are regulated and determined by the authorities itself based adopted methodologies. Therefore, it is the Municipality Council and the Mayor which is responsible for setting up the prices of utility services within the utility companies in BiH, while the company's management provides price policy suggestions but is not in the position to enforce any tariff changes. The full responsibility of the tariff policies in BiH is at the municipal/cantonal level. There is no entity or state level legislation, regulation body or whatsoever, which regulates or prescribes a common methodology on thermal energy and water prices for district heating/water utility companies. Due to the situation where artificially prices are kept low, annual financial losses are the very usual reflection among BiH utilities resulting in cross-subsidizing mechanisms – covering financial losses through the local budget.

Electricity prices for public consumers and households are the lowest in Europe (in accordance to Eurostat only Macedonia has lower electricity prices) valuing on average 2.5 times less per kWh consumed. Furthermore, there are no fossil fuel taxes or emission limitations for public and residential buildings in place which would stimulate energy efficiency and renewable energy investments.

As a result of non-existing market-driven energy prices and the existing market-driven EE / RES equipment and technology purchase prices (equivalent to average European prices), the average BiH citizen is not motivated to invest into energy efficiency improvement measures or the utilization of renewable energy sources due to mid and long-term expected pay off periods. Moreover, there are no financial mechanisms or incentives in place for the general public/individuals which might act as encouraging for such investments. In the public sector in Bosnia and Herzegovina, within the last five years, several financing projects (mostly grant and/or co-financing) contributed to the implementation of energy efficiency measures. However, all of them are rather small scale in terms of volume of funds available and/or in terms of distribution of funds. Moreover, the 1st NEEAP, although not adopted, reflects a strategic approach to achieve EE targets of FBiH and RS by 2018 and proposes financial incentives for residential, public and commercial buildings. Although foreseen within the NEEAP, currently there are no financial incentives (VAT reduction on selected EE products, Tax rebates for EE investments, soft loan supported by government/authorities) for energy efficiency products, activities or measures in Bosnia and Herzegovina. The noteworthy concern is that once the primary and secondary legations are in place and adopted in both entities, and the monitoring and reporting mechanisms are developed and in position to replace ad-hoc for strategic EE / RES

[10] World Bank data

investment decision making, the required financial market mechanisms and modalities will not be developed to a stage to be able to support investments into BiH's energy saving targets and generate significant employment. Therefore, it is highly recommended that activities on the development of financial mechanisms and financial modality enabling legislation take place in parallel to the development of monitoring and reporting mechanisms.

2.3. SOLUTIONS - EE / RES AS ECONOMY AND LOCAL DEVELOPMENT DRIVING FORCES IN BIH

The currently developed secondary legislation on EE and RES, which shall be adopted within FBiH and RS governments, will not be sufficient to enforce and implement energy related EU Directives in Bosnia and Herzegovina resulting in continued failure to meet the country's obligations to the ECT. Moreover, while secondary legislation and action plans will and do mention various types of possible financial EE / RES mechanisms, the legal framework within the entities and state level for financial mechanisms has not yet been analysed nor required legal framework amendments identified.
This situation analysis suggests activities to be undertaken by the UN system in BiH and the international community in order to overcome the barriers on transposing and implementing energy efficiency and renewable energy related directives into BiH legal environment and thus creating a favourable investing environment resulting in economic and local development through reallocation of public budget funds (from energy/water expenses to investments in education, health, infrastructure, culture etc.) and generation of employment.

In order to enable such an environment, further focus should be given to the following matters:

- Bottom-up approach to energy management capacity building and legal obligation understanding
- Creating monitoring and reporting mechanisms
- Identification and development of EE / RES financial mechanisms
- Further policy level interventions

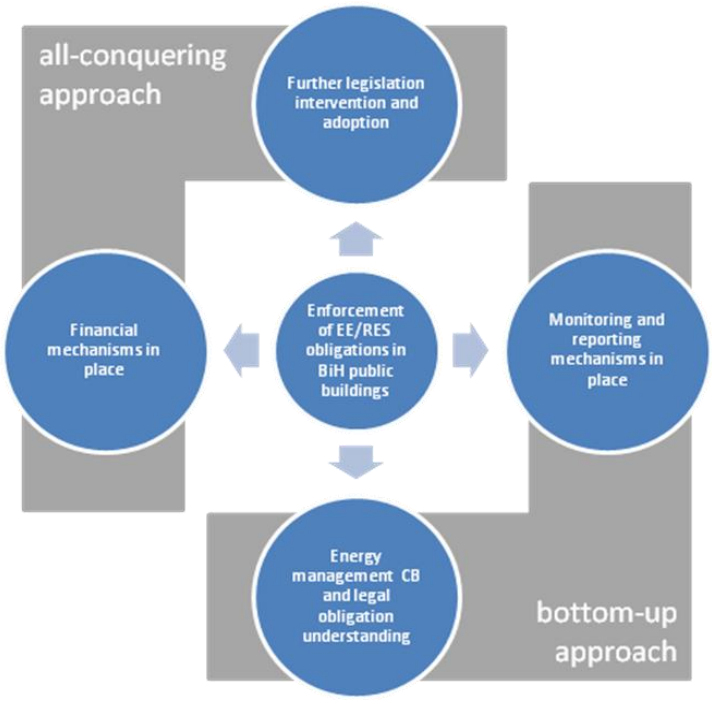


Figure 1 - EE / RES enabling investment environment model in BiH[11]

In accordance to UNDP experience, and due to targeted implementation requirements and the highly fragmented public sector jurisdictions, a bottom-up approach for energy management and legal obligation understanding, as well as the development of a sustainable monitoring and reporting mechanisms, is essential. A comprehensive, all-conquering approach with much stronger involvement of entity and cantonal line ministries is required for further EE/ RES policy level interventions and the alignment of EU directives to BiH environment, as well as for the identification and development of EE /RES financial mechanisms to support the public sector in its energy conservation activities.

2.3.1. Bottom-up approach to energy management capacity building and legal obligation understanding

If EU directives and primary and secondary legislation on energy efficiency and renewable energy are expected to be enforced in BiH, it is essential that lower level governments understand BiH's obligations on targeted energy savings, in particular goals set for public buildings, and develop energy management decision making capacities within their jurisdictions. In the BiH given complex administrative context, only if municipalities, and especially cantonal and entity line ministries, have an understanding and are enabled to manage energy related issues, Bosnia and Herzegovina will be able to fulfill its multilateral obligations and create sustainability of required EE / RES undertakings in the public sector.

Therefore, UNDP's *Municipal Training System* and *Green Environmental Development* projects already target the municipal and cantonal/ministry level authorities with the aim to familiarize them

[11] Base on: Energy efficiency in BiH - current status, legal regulations, implemented and planned projects, S. Avdić, 2012

with required energy management knowledge and skills and create understanding of BiH's current legal obligations and upcoming cantonal/municipal responsibilities in the implementation chain.

2.3.2. Creating monitoring and reporting mechanisms

Given the lack of entity and state level statistical data, as well as unadjusted energy related indicators and parameters required for systematic energy management decision making and reporting, it is obvious that improvement of the current countrywide organisational set-up and implementation arrangements in the area of energy monitoring database management and reporting is one of the key energy related challenges in Bosnia and Herzegovina. Unlikely the environmental monitoring in BiH, where a huge number of institutions collect data on various environment indicators, energy consumption and energy emission indicators from public buildings are not recorded, monitored or analyzed in a organizational, structured way at all. Moreover, most of the ministries responsible for different type of public buildings do not have an overview of their building stock basic parameters (square meters, type of fuel, energy and water related costs etc.).

In these circumstances, a top-down approach to track energy consumption, emissions and achieved energy savings is not reliable and completely inconsistence with reporting requirements. Therefore, entity authorities e.g. Environmental Funds (since identified and required by Law on EE as institutions supporting the EE targets), should be involved in the data collection process. In order to help to overcome these challenges, to provide authorizes with a reliable monitoring tool through which they can base their further EE / RES investment decision on, and to contribute to BiH's reporting obligations to multilateral agreements (ECT, UNFCCC), UNDP's is guiding its efforts to develop a clearly structured and unified energy and CO_2 emission monitoring and reporting system in both entities in BiH. Namely, UNDP invested 1.000.000 USD into its in-house developed Energy Management Information System – EMIS, which shall provide BiH's authorities the operational component for energy and emission indicator (consumption, costs etc.), target setting and achieving, management and reporting monitoring tool. Moreover, it will enable state, entity and cantonal line ministries/institutions (e.g. key focal points - Ministry of Foreign Trade and Economic Relations of BiH, cantonal ministries of Spatial Planning and Environment, Ministry of Spatial Planning, Civil Engineering and Ecology of RS, Environmental Funds of FBIH and RS) to develop their baseline building stock and statistical database of public buildings under their jurisdiction.

Through UNDP past activities on the *MGD-F Environment* project, EMIS has been installed in about 500 buildings throughout BiH in which, among other parameters, energy consumption, energy costs and CO_2 emission are monitored on a monthly basis. Until now more than 1.000 public sector buildings are in the database. Moreover, EMIS data collection sets the basis for the creation of Energy Efficiency Action Plan's on Municipal/Cantonal/Entity level – one of the EE legislation requirements. The main challenge however, is to adopt energy and emission monitoring indicators

and reporting mechanisms on cantonal government/ministry level in FBiH and RS, which will assure a long-term sustainable and reliable mechanism in place.

UNDP is currently under the *Green Environmental Development* project aiming to create and set in place an energy monitoring and reporting mechanism for public buildings in two Cantons in FBiH – Una-Sana Canton and Tuzla Canton (key focal points - cantonal ministries of Spatial Planning and Environment), as well as one Ministry in RS – Ministry of Education and Culture. Once in place, the developed mechanism could be replicated to other ministries and cantons, while creating a reporting flow on required data to entity governments (through Environmental Funds, since identified as the key institutions for databases in accordance to Law on EE) and further to state level MoFTER. The solution to BiH's non-existing energy reporting and monitoring mechanisms is a top-down target setting flow (state-entity-canton/ministry/municipality), assured through entity adoption of primary and secondary legislation on EE / RES, and a bottom-up reporting flow (canton/ministry/municipality-entity-state). Further inter-entity and entity-state cooperation and coordination shall follow.

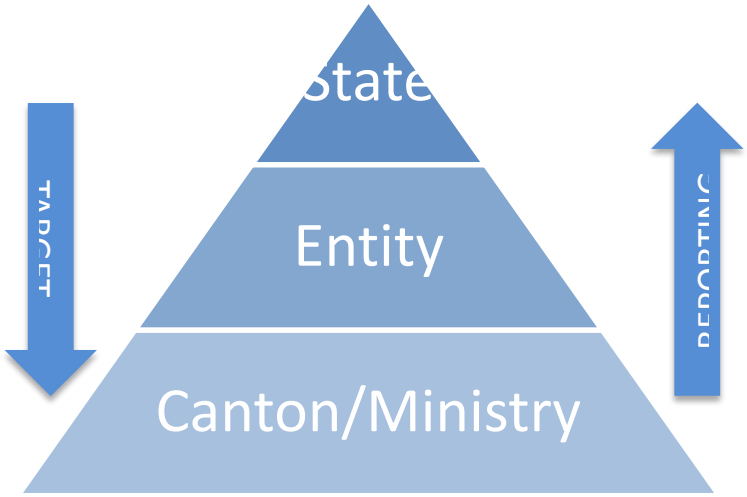


Figure 2 – Proposed BiH's reporting and monitoring mechanisms on energy consumption, energy costs and CO_2 emission

One of the most important issues regarding energy monitoring in BiH is to avoid a huge number of institutions collecting data, without coordination and policy guidance. Otherwise, it would not ensure data compatibility, or take each other's practices into account, when upgrading or further developing systems for data collection and management.

2.3.3. Identification and development of financial mechanisms

Once the legal framework for EE / RES primary and secondary legations is in place, energy efficiency improvement activities and the utilization of renewables (and thus the generation of employment and economic development) will not occur in volume required for meeting BiH's targets due to non-existence of financial mechanisms which have to contribute to its implementation. Namely, although some financial supporting mechanisms and models for EE / RES investments in the public sector have been listed by non-official NEEAP, a clear identification of necessary legislative bylaws, required amendments to existing legislation, identified actors, obligation and responsibilities, as well as financing models, incentives and mechanisms suitable for BiH's environment are not identified nor is it discusses or on the political/authority agenda present. This vacuum has to be over-bridged with clear directions and guidelines for compulsory actions leading to the formation of sustainable and reliable financial market mechanisms and modalities in BiH's entity level institutions (ministries, development banks, funds etc.).

UNDP's past activities contributed to draft amendments to Law on Environmental Protection Fund in RS, FBiH and Brčko District, with the intention to extend their jurisdiction to energy efficiency and thus enable them to provide constant EE project financing through their existing financial mechanisms. In RS amendments have been adopted, while in FBiH they are in adoption phase. Currently, UNDP's *Green Environmental Development* project is developing a Study on development of energy efficiency financial mechanisms (revolving fund, performance based granting, soft-loans, guarantees) within Environmental Protection Fund in FBiH and RS, which is on-going and expected to be finished by mid-August for FBiH and by end of October for RS. Some finding of the Study for FBiH are showed in chapter 4.3.

2.3.4. Further policy level interventions

Further policy level interventions (secondary legislation related to technical parameters and requirements), due to its partial drafting and not fully alignment to BiH context from previous projects, should be developed and adopted in order to fully implement BiH EU accession requirements and enable a EE / RES market in BiH. Further policy development shall take place for:

- EE / RES in buildings sector
- EE of appliances
- EE / RES in industry sector
- EE / RES market mechanisms

III. GREEN ECONOMIC DEVELOPMENT PROJECT

3.1. BASELINE SITUATION / ENERGY EFFICIENCY AND FINANCIAL MECHANISMS IN BIH

The currently developed legislation on energy efficiency (EE), which shall be/already is adopted within FBiH and RS governments, will not be sufficient to enforce and implement energy related EU Directives in Bosnia and Herzegovina. Moreover, while secondary legislation and action plans mention various types of possible financial EE mechanisms, the legal framework within the entities and state level for financial mechanisms has not yet been analyzed nor required legal framework amendments identified. Therefore, in order to enable an EE investing environment, further focus should be given to the following matters:

- Bottom-up approach to energy management capacity building and legal obligation understanding
- Creating monitoring and reporting mechanisms on energy consumption and energy costs
- Identification and development of EE financial mechanisms within Environmental Funds

Required interventions / suggested next steps

In accordance to UNDP experience, and due to targeted implementation requirements and the highly fragmented public sector jurisdictions, a bottom-up approach for energy management and legal obligation understanding, as well as the development of self-sustainable monitoring and reporting mechanisms, is essential. Moreover, with the aim to create a favorable investing environment resulting in economic and local development through reallocation of public budget funds (from energy/water expenses to investments in education, health, infrastructure, culture and further energy efficiency projects etc.) and generation of employment, the identification and adoption of EE financial mechanisms to support the public sector in its energy conservation activities is required. UNDP's Green Economic Development project addresses those aspects required to assure long-term project goal sustainability.

National and sub-national alignment of GED

In regards to its obligation to the Energy Community Treaty, which obligates BiH to transpose, adopt and implemented various energy related EU directives in its national legal framework, the Law on Energy Efficiency has been drafted and adopted in the entity of RS and is expected soon to be adopted FBiH (by August 2014). As per Law on Energy Efficiency, Environmental Protection Funds will be financial institutions supporting the enforcement of EE Law and EE targets and will partly also act as EE Agencies (IT solutions/systems for Law on EE will be placed and administrated by Environmental Funds) and thus, once their capacities and infrastructure is developed, will be able to provide BiH state Ministry MoFTER with required reliable and valid data on energy consumption, actual energy savings, energy indicators and other required indicators. Doing so, BiH will be able to fulfill its reporting and coordinating obligations in accordance to EC's Energy Community Treaty on energy consumption monitoring and energy savings achievements, as well as on other data.

GED project activities will enable entity Environmental Protection Funds to monitor, analyze and evaluate energy consumption, costs, emission and energy savings data from public sector buildings, and provide the state of BiH (MoFTER) with consolidated reports for further submission to Energy Community Secretariat and other multilateral agreements (for example – United Nation Framework Convention on Climate Change). UNDP's Energy management Information System (EMIS – energy monitoring tool for public sector buildings) will be part of Law on EE defined IT solutions.

3.2. UNDP'S PAST EXPERIENCE AND ON-GOING ACTIVITIES

Since 2009, UNDP CO BiH is one of the leading agencies promoting energy efficiency in Bosnia and Herzegovina, primarily acting within MDG-F Environment and Climate Change programme (ended in May 2013). Focus of UNDP's work was on piloting introduction of energy efficiency in the public sector across Bosnia and Herzegovina by implementation of replicable energy efficiency projects and introduction of Energy Management Information System (EMIS – energy, costs and emission monitoring software with a vast number of analytical features) into public sector buildings.

During the previous five years, a total number of 38 energy efficiency pilot projects (small to medium scale) were implemented across the country with following results:

- Investment amount: 4,2m USD
- Co-financing amount (end users and responsible institutions): 1,2 m USD
- Total energy cost savings: 700,000 USD annually (up to six years average pay-back period)
- Total emission reduction: 2.200 t CO2 annually
- Total direct employment achieved: 664 man/months (total worth in salaries 485,000 USD)

The project included:

- Implementation of energy conservation and renewable energy measures in existing public sector buildings (thermal-insulation, EE windows, efficient heating systems and heat distribution)
- Fuel switch projects (fossil fuel to biomass fueled boilers)
- Automatized energy consumption regulation and management of public sector buildings
- Implementation of EE public lightning
- Education on energy management of public sector buildings for end-users

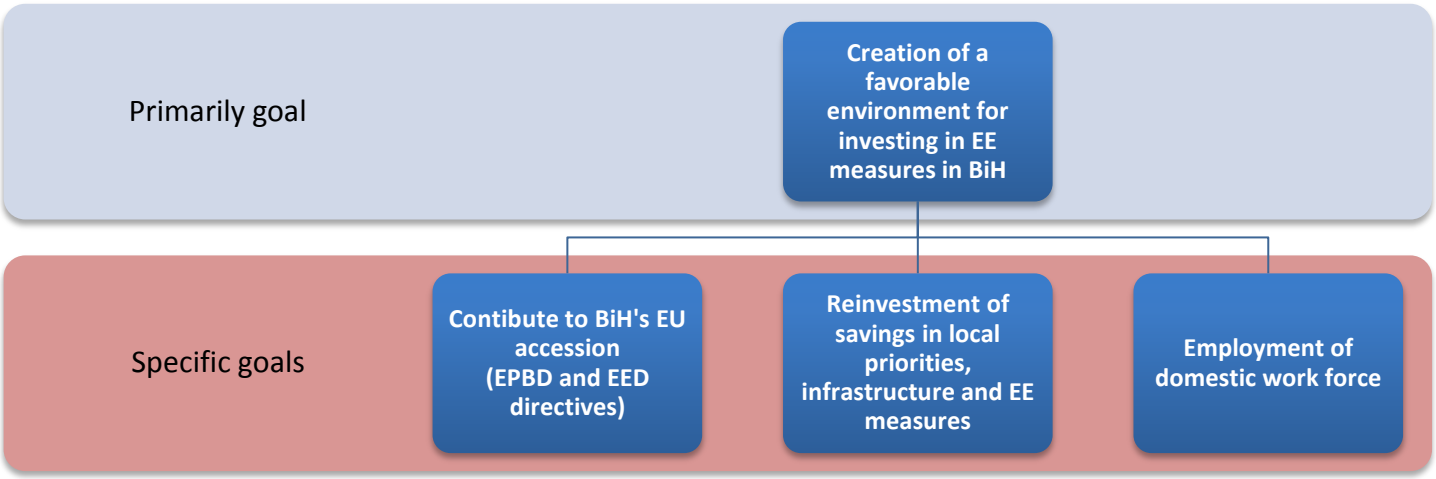
Currently, as follow up activities, under the Green Economic Development project (GED), UNDP Bosnia and Herzegovina and its locally co-financing/supporting project partners (Environmental Protection Fund of FBiH, Environmental Protection and Energy Efficiency Fund of RS, cantonal governments in FBiH and ministries in RS) aim to institutionalize energy management within the public sector buildings in Bosnia and Herzegovina and to create decision making processes which lead to the implementation of energy efficiency infrastructure projects.

3.3. GREEN ECONOMIC DEVELOPMENT – GED PROJECT

The primarily goal of **Green Economic Development** project is to contribute to the creation of a favorable environment for investing in EE measures in BiH.

Within UNDP's GED project, a five year Programme (2014 - 2018) has been launched with Environmental Protection Funds in BiH, and supported by a number of cantonal governments in FBiH and ministries in RS. The aim of the Programme is to:

i. develop capacity and skills of Environmental Funds (systematic approach to decision making processes of energy efficiency investments)
ii. develop and adopt sustainable financial mechanisms within Environmental Protection Funds for EE project financing (performance based granting, revolving mechanisms, soft-loans, etc.)
iii. institutionalize energy monitoring and reporting mechanisms / energy management
iv. achieve budget cost savings through implementation of EE projects in public sector buildings and reinvest savings into energy efficiency or other infrastructure projects
v. generate employment of domestic workforce
vi. increase public awareness and understanding of EE benefits
vii. contribute to BiH's EU accession (EPBD and EED directives)



3.3.1. Overview of GED Programme Components

In accordance with the goal, the Programme consists of five interlinked and supplementing project components:

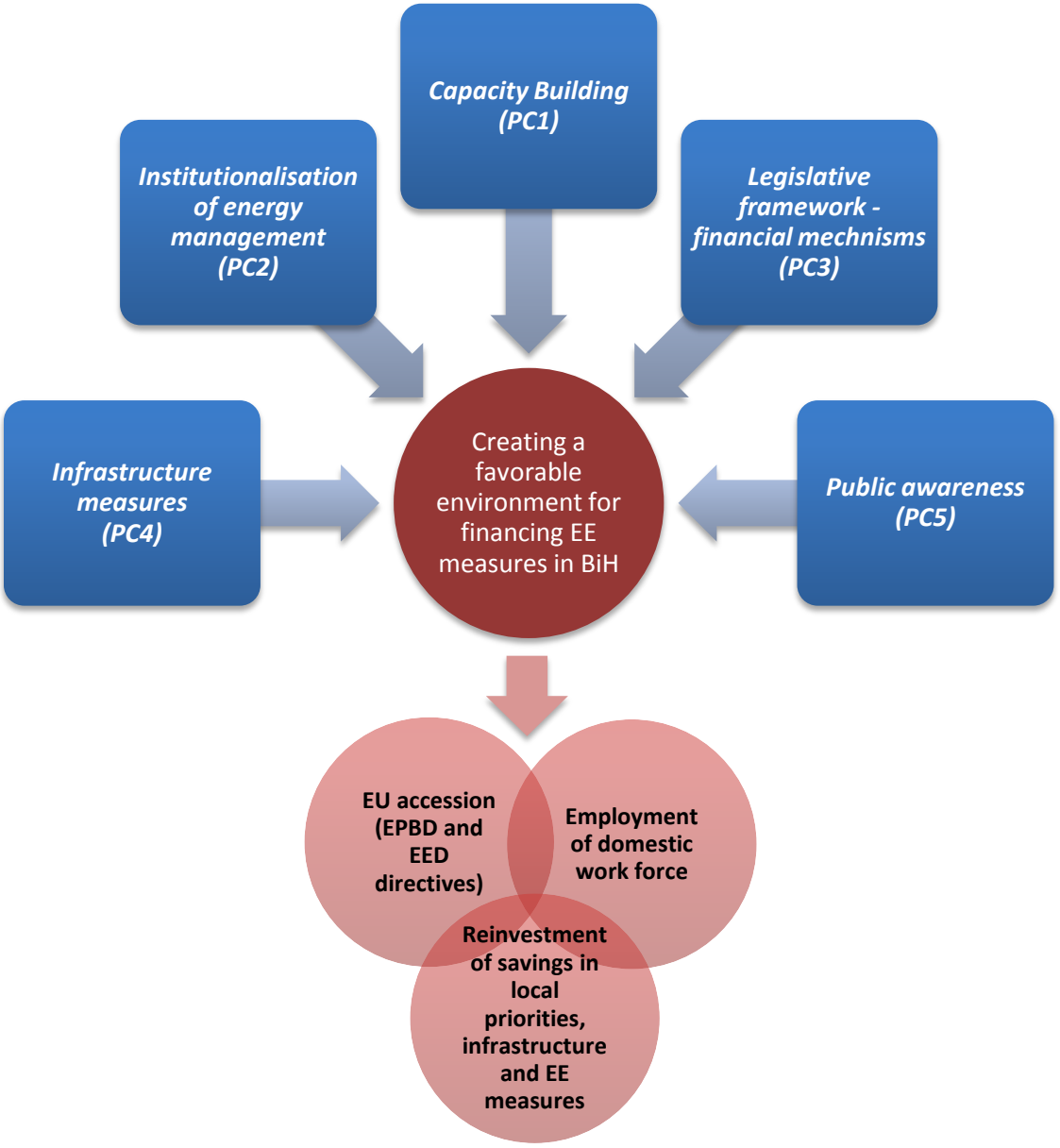
I. ***Capacity Building (PC1)*** – Development and capacity building of Environmental Funds and energy professionals.

II. ***Institutionalization of energy management (PC2)*** – Institutionalization of energy, costs and emission management and monitoring in public sector buildings of BiH.

III. ***Legislative framework / financial mechanisms (PC3)*** – Development and adoption of sustainable financial mechanisms within Environmental Protection Funds.

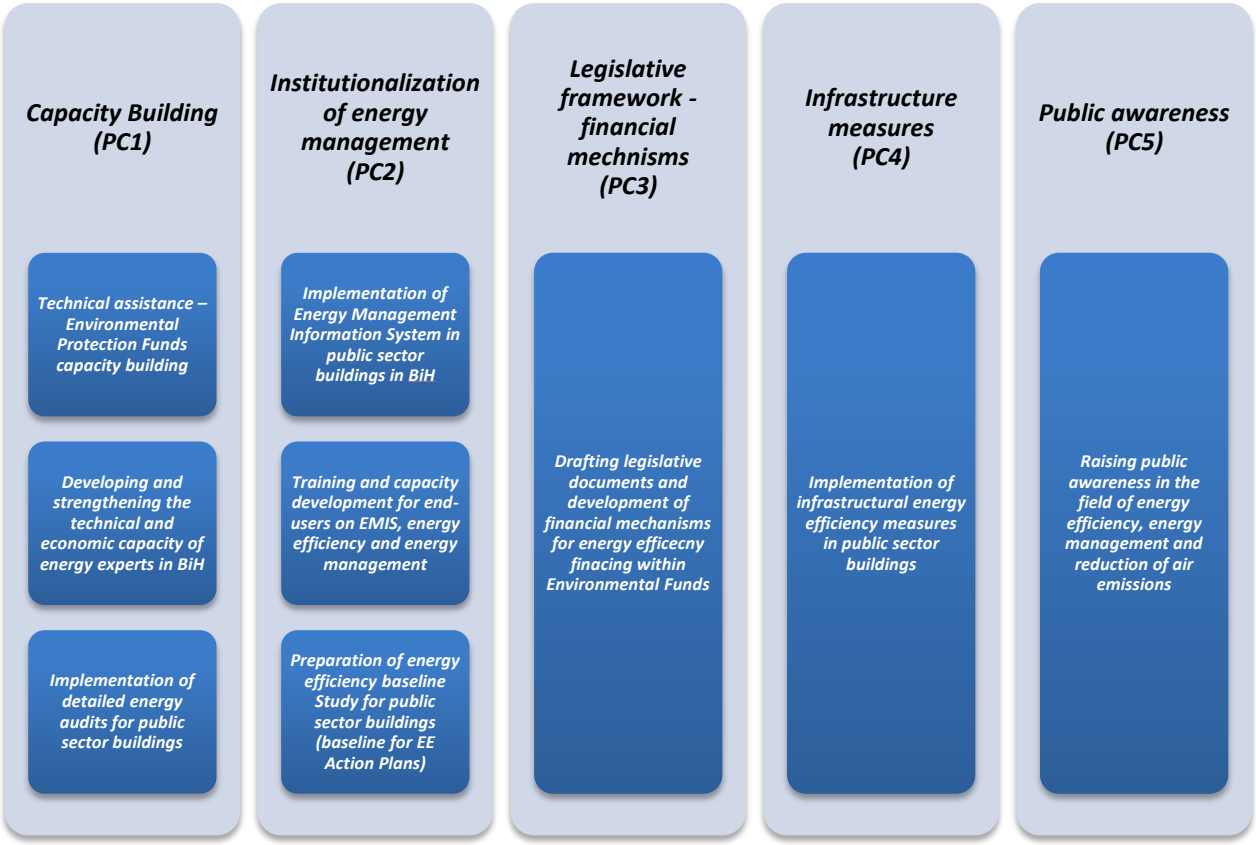
IV. ***Infrastructure measures (PC4)*** – Implementation of energy efficiency measures in public buildings.

V. ***Public awareness / marketing campaign (PC5)*** – increasing public awareness on energy efficiency



3.3.2. Overview of GED planned activities

GED activities can be grouped into nine activity groups, with strong correlation and interdependency among those activities and five Project Components. Short overview and description of activities is given below.



- ***Technical assistance – Environmental Protection Funds capacity building (Project Component 1)***

  UNDP staff providing technical assistance to the Environmental Protection Fund in the field of energy efficiency, energy management and management of EMIS, in the form of consultative, advisory and technical support, as well as hiring consultants for niche expertise/knowledge.

- ***Developing and strengthening the technical and economic capacity of energy experts in BiH (Project Component 1)***

  Developing and strengthening the capacity of energy experts, and create understanding of the legislative framework in FBiH/RS/BiH on EU directives in the field of energy efficiency and energy management. Improvement of techno-economic skills necessary to perform energy audits and making bankable documents will be made through the realization of capacity building activities via training programs and the establishment of the Association of Energy Professionals in BiH.

- ***Implementation of detailed energy audits for public sector buildings (Project Component 1)***

  Based on the collected energy, economic and emission data through EMIS, and the resulting indicators, a techno-economic prioritization will be conducted and about 100 buildings during project implementation will be chosen to conduct detailed energy audits. Capacity building and skills development of Funds internal capacities on energy efficiency investments decision making processes will be ensured through these evaluation, analysis and prioritization activities.

- ***Implementation of Energy Management Information System in 5.000 public buildings in BiH (Project Component 2)***

  UNDP BiH and the Environmental Protection Funds in joint activities will contribute to the implementation of Energy Management Information System (EMIS) in more than 5.000 public buildings in BiH (3.000 in FBiH and 2.000 in RS).

- ***Training and capacity development for end-users on EMIS, energy efficiency and energy management (Project Component 2)***

  Training for end-users on EMIS, energy efficiency / energy management in public buildings for 5.000 identified public sector buildings end users.

- ***Preparation of energy efficiency baseline Study for public sector buildings (Project Component 2)***

  The baseline Study on the energy efficiency of public buildings in BiH is complementary to the implementation of EMIS. The study complements EMIS in terms of investment calculations of energy efficiency measures and their environmental effects. Using the methodology of "walk-through audits", 500 buildings per year in FBiH will be processed and analyzed through a single Study that in addition of individual assessment for each facility will provide an analysis of all the buildings together, according to the type and sector (educational institutions, health care, etc.). This baseline Studies will be used as baseline data for EE Action Plans.

- ***Drafting legislative documents and development of financial mechanisms for energy efficiency financing within Environmental Funds***
  (Project Component 3)

  Drafting legislative and guidance documents on financing energy efficiency in BiH includes the creation of guideline documents and secondary legislation, in order to create a sustainable financial model for the establishment of financial mechanisms for energy efficiency financing within Environmental Protection Funds. The financial mechanisms will include revolving mechanisms, performance based granting, soft loans, guarantees and other financial modalities which could be provided by the Environmental Funds with the aim to create a legal framework for sustainable investments and reinvesting possibilities in energy efficiency.

- ***Implementation of infrastructural energy efficiency measures in public sector buildings (Project Component 4)***

  On the basis of detailed energy audits findings, infrastructure measures will be implemented in up to five public sector buildings per year in order to conduct so called "Deep retrofits" (in accordance with the requirements of EU directives EED and EPBD) and respecting newly developed and adopted construction and energy conservation standards (EPBD directive).

- ***Raising public awareness in the field of energy efficiency, energy management and reduction of emissions to air***
  (Project Component 5)

  A comprehensive marketing campaign including TV spots, radio jingles and social network will be developed and implemented through the Programme duration. Raising public awareness events in 15 BiH cities are planned to be conducted. Designing, printing and distribution of educational publications that explain EE from the basics to the technical details related to energy savings, energy management and emission reduction. Publications would be intended for a wider audience - from young people, who are first confronted with these issues, to professionals working in the energy sector. In addition, the development of an informative web portal on energy efficiency, energy management and reducing air emissions will be conducted.

3.3.3. Description of GED planned activities

Activity 1 - Technical assistance to Environmental Protection Funds, EE capacity building and skills development (refers to PC1)

- Since energy efficiency approach on projects is still new and undeveloped in BiH, UNDP staff will provide technical assistance to the Environmental Protection Funds in the field of energy efficiency, energy management and management of EMIS (*Energy Management Information System*), understanding of Laws, by-laws and EU directives. Assistance will be conducted in the form of consultative, advisory and technical support, through annual training for Fund staff. Also UNDP staff will provide technical assistance throughout project implementation and skills development via on the job training/assistance on all phases of project:
    - Analysis, evaluation and development of energy indicators and reports out of EMIS database
    - Analysis, evaluation and development of prioritization list of public sector buildings for energy audits
    - Analysis, evaluation and development of technical, economic, financial, environmental and social parameters/factors (including co-financing from end-users) and development of ranking list of public sector buildings for the implementation of EE infrastructure measures
    - Monitoring and verification of energy and costs savings, emission reduction
    - Tracking of indicators
    - decision making process cycle/methodology for EE infrastructure investments (identification, evaluation, decision, implementation and monitoring/reporting)
- Assistance and trainings are very important as a first step, since they will provide understanding of energy savings potential, improve knowledge and understanding of relevant EU directives, Laws and by-laws regarding energy management. This also provides practical help in understanding energy indicators, decision making process and evaluation of public sector buildings for the implementation of EE infrastructure measures
- This activity correlates with all others, since it upgrades knowledge and gives technical assistance through all phases of the project assuring that all other activities can be implemented smoothly and timely.

Activity 2 - Developing and strengthening the technical and economic capacity of energy experts in BiH (refers to PC1)

- In order to assure high quality and reliable project documentation (energy audits - rightly chosen and evaluated measures, cost-optimal scenarios for implementation of EE investments), and the creation of a competitive market in the field of energy efficiency, developing and strengthening the capacity of energy experts is necessary, as well as establishing Association of Energy Professionals in BiH. The main objective of the association is to secure the best legislative and regulatory framework for a professional approach to thermal engineering standards, practices, education and development in the BiH. UNDP will also give support in developing and facilitating a *Professional Development Programme for*

energy professionals in order to create further in-depth understanding and increase skills of energy professionals in BiH in the area of energy efficiency, renewables, energy management, relevant energy policies and norms and standards. UNDP will develop ToRs for delivery of both, trainings (Professional Development Programme) and establishment of the Association, will be in accordance with UNDP technical requirements and procedures (SOPs).

- Since energy auditing is a multidisciplinary activity, capacity building for energy professionals (target group - mechanical, electrical, civil engineers and architects) will be provided to ensure high quality and reliable development of detailed energy audits on the BiH market with incorporated "deep-retrofitting" EE measures. Moreover, the understanding of economics and financials is one of key outputs of detailed energy audits based on which investment decision will be made. Therefore, the CB of energy professionals will also focus on creation of understanding and skills of energy professional on economics and project financing. The further development of mentioned skills and capabilities of energy professionals will result in higher quality project ideas and assessments of EE projects which could be financed by other donors, projects or private investors. The *Professional Development Programme for energy professionals* will also be used as a platform to communicate to energy professional GED project's objectives, results achieved and lessons learned.
- The *Professional Development Programme for energy professionals* will be open to all energy professionals. UNDP will develop ToRs for development and delivery of *Professional Development Programme for energy professionals*, in accordance with internal procedures and will tender the conductance and development of energy audits. It is envisaged that senior energy experts from BiH and the region provide lectures to energy professionals on four two-day events in BiH. In later GED project stage, the *Professional Development Programme for energy professionals* could focus on other development areas for energy professionals or could be transferred to another target group (depending on indicator progress and decision of Project Board).
- Following tasks will be conducted (developed by UNDP and delivered by contractor):
    i) Delivery of training activities and materials for each participant. Hold four events in four different sites in BiH which will last for two days each. Each event/ program will have the same content and will take place at other locations and on other dates.
    ii) Each event/training program consists of two Professional groups (Professional group 1 for architecture and civil engineers, Professional group 2 for mechanical and electrical engineers and one Combined Professional group for all participants for common introductory presentations, policy, rulebooks etc.)



iii) Expected topics to covered, but not limited to:

Combined Group:

Common introductory presentations, policy, rulebooks etc. for all participants such as, but not limited to:

- Law and Technical regulations in the field of energy efficiency, thermal protection, sound protection etc.
- Law and rulebooks on construction products
- Fire protection through laws and regulations – Application
- Etc.

Professional Group 1:

Presentations of various topics in the field of architecture and civil engineering such as, but not limited to:

- Restoration of old buildings in terms of increasing energy efficiency and energy conservation
- Performance of construction materials in the fire environment - reaction to fire
- Closing the openings in fire-resistant structures
- Interventions, reconstruction and adaptation of the buildings under protection
- Improving the sound insulation in new buildings and existing buildings
- Low energy and passive houses
- Mounting and installation of windows in accordance to the energy savings (EnEv) Standard - RAL installation instructions
- Thermography in Buildings

Professional Group 2:

Presentations of various topics in the field of mechanical and electrical engineering such as, but not limited to:

- Air systems balancing and automation systems in order to take advantage of energy efficiency
- Alternative systems and renewable energy sources (focus on biomass and solar)
- Heat pumps for heating, cooling, ventilation and air conditioning
- Heat recovery

- New advances in the fire protection systems to spread through the ventilation and air conditioning systems
- Central monitoring and control system (CCMS) and Building Management Systems (BMS)
- Microclimate conditions in industrial facilities
- Energy efficient lighting
- Management of public lighting
- Cogeneration and trigeneration plants - electricity, heat and cooling energy
- Photovoltaic power stations and electric power generation

The above mentioned key topics are expected to be covered. However, this list is only indicative and the consultant is encouraged to suggest further/alternate topics, which can help in achieving the objectives. Each event at the Professional Development Program should have about 40 participants.

iv) Evaluate the professional development program/participants (energy professionals) using a feedback form on change of in-depth understanding of energy efficiency and renewable techniques and technology, ability to conduct detailed energy audits with "deep-retrofitting" measures after each event.

- Since the right energy efficiency approach is still undeveloped within in BiH (low understanding on Law requirements and obligations, standards, and cost-effectiveness by energy experts), skills development and capacity building of energy experts, as well as the establishment of Association of Energy Professionals in BiH (in order to be able to tackle and enforce standards and ensure respect to professional approach to EE and other thermal energy related issues) is required.

- This activity contributes to a successful functioning of energy efficiency market, and for high quality and reliable implementation of infrastructural measures.

Activity 3 - Drafting of detailed energy audits for public sector buildings (refers to PC1):

- A prioritization list of public sector buildings for the conductance of detailed energy audits, based on technical and economic parameters, collected energy consumption and costs data, as well as energy conservation potential and emission reduction emission potential (all based on EMIS database and analysis) will be developed by UNDP and Fund staff. At least 100 buildings during project implementation will be chosen to conduct detailed energy audits. UNDP will develop ToRs for delivery of energy audits, in accordance with internal procedures and will tender the conductance and development of energy audits. DEA (detailed energy audits) will provide an in depth analysis of the current baseline situation of a public sector building, determine energy losses, current energy efficiency of the heating, cooling and lightning system (including all other energy forms and/or energy usage

purposes). It will analyze and propose possible measures of increasing energy efficiency of the building and perform a comparative analysis of different Scenarios (combination of measures), based on technical, economic and environmental related indicators.

- Capacity building and skills development of Funds internal capacities, on energy efficiency investments decision making processes, will be ensured with UNDP coordination and on the job training through evaluation, analysis and prioritization activities. Finally, a ranking list of public sector buildings for the implementation of EE infrastructure measures will be developed and updated annually.

- For enabling this Activity it is necessary to establish EMIS database through Activity 4. This Activity represents practical usage of earlier acquired knowledge, as well as developing and upgrading skills for the Funds future work (Activity 1).

- As specified before, this is a complex process which can be showed as follows:



- Evaluation process and development of priorities/rankings are necessary for choosing the most cost-effective investments, as well as to justify the investment decision and to assure transparency throughout the process. As a start of promoting energy efficiency, public sector buildings are great example to show how both, comfort and savings could be achieved.

Activity 4 - Implementation of Energy Management Information System in 5.000 public sector buildings in BiH (refers to PC2):

- Easiest way of monitoring, evaluating and making analysis of energy consumption is using one single application. UNDP BiH and the Environmental Protection Funds in joint activities will contribute to the implementation of Energy Management Information System (EMIS) in more than 5.000 public buildings in BiH (3.000 in FBiH and 2.000 in RS). This will provide clear view of energy consumption for public buildings, as well as possibilities of savings. Fund and UNDP will extend this database through annual Call for Proposals for identification of public sector buildings/end-users interested for EE investments, as well as through sub-national partner involved in GED.

- Public sector buildings submit application to Fund/UNDP by filling out pre-defined survey of interest with basic building data and co-financing potential, followed by the creation of

accounts (static input data) for the public buildings. Further input/information on consumption and costs of energy and water in the last 36 months period is collected and entered into EMIS. During this process UNDP staff will provide technical support and assistance to contact persons in identified public facilities. Besides that, UNDP staff is preparing technical, economic and environmental parameters and energy conservation potential indicators within EMIS database, managing, administrating, keeping maintenance, monitoring, error identification and correction activities for further development of EMIS database. EMIS provides the ability for monitoring and reporting of achieved energy and cost savings (kWh, KM), CO_2 emission (t CO2) reduction and other specific indicators.

- By using a software which enables a continuous monitoring of energy consumption and costs (monitor parameters before/after EE measures implementation) additional energy savings can be achieved (for example, it has been proved in public sector buildings in Croatia that just using EMIS brings energy consumption down up to 20%, since the awareness of users is improved, and they know that every usage is monitored).

- By developing a database of public sector buildings it is possible to accomplish Activity 3 (and choose the public sector buildings with the highest energy conservation potential for conducting detailed energy audits) and to monitor parameters before/after measures implementation, which is possible only if public building representatives pass the training (Activity 5) and keep updating information about energy consumption (monitoring and reporting is part of obligation in accordance to Law on EE, while EMIS provides the tool for monitoring energy consumption in public sector buildings).

Activity 5 - Training and capacity development for end-users on EMIS, energy efficiency and energy management by UNDP (refers to PC2):

- UNDP staff will provide training for end-users on EMIS, energy efficiency / energy management in public buildings for abovementioned 5.000 identified public sector buildings end-users. UNDP will organize training facilities throughout BiH, equipment (laptops) and travel arrangements, contact all identified end-users, provide information, conduct training on EMIS and provide technical support and assistance to trained persons throughout project implementation.

- This database will provide sub-national governments with the ability to monitor and report on energy consumption, energy savings and reduced CO2 emission from public sector buildings e.g. in accordance and obligation with Law on EE and defined EEAPs. Moreover, the Environmental Funds are envisaged per Law to enable and conduct reporting activities through IT systems.

Activity 6 - Preparation of energy efficiency baseline Study for public sector buildings and EEAPs (refers to PC2):

- The baseline Study on energy efficiency of public buildings in BiH is complementary to the implementation of EMIS. The study complements EMIS in terms of investment calculations of energy efficiency measures and their environmental effects. Using the methodology of "walk-through audits" up to 1000 buildings per year will be processed and analyzed through a single Study that in addition of individual assessment for each facility will provide a combined analysis of all the buildings, according to the type and sector (educational institutions, health care, administration, cultural etc.). This baseline Studies will be used as baseline data for EE Action Plans. UNDP will, in accordance with UNDP technical requirements and procedures (SOPs), develop ToR for the development of energy efficiency baseline Study for public sector buildings on annual basis, and ToR for the assistance to development of Energy Efficiency Action Plans and Reporting Mechanisms (EEAP/RM) for sub-national governments, and coordinate and manage the development of both, Study and EEAP/RM. The further step is the adoption of EEAP/RM by sub-national governments which are partners within the GED project and which are obligated to do so as per Law on EE.

- This activity will set the base for EEAP plan to be planned, enforced and implemented, and will enable sub-national governments to seek for EE investment support by Environmental Funds in the upcoming (post GED project) period.

Activity 7 - Drafting legislative documents, development of financial mechanisms for energy efficiency financing within Environmental Funds (refers to PC3):

- As the first step within this activity, a Study focusing on the further development of energy efficiency based financial mechanisms and modalities (grant, revolving fund, loans, guarantees etc.), as well as required legislative development/changes within Funds, will be developed. Based on the findings and Fund's senior management decision on further development, UNDP will support the development of additional required internal acts and legislative documents to enable a fully operational grant, EE Revolving Fund, loans, performance based granting etc. The financial mechanisms will include revolving mechanisms, performance based granting, soft loans, guarantees and other financial modalities which could be provided by the Environmental Funds with the aim to create a legal framework for sustainable investments and reinvesting possibilities in energy efficiency. Technical assistance and ToRs for developing Study on improving and developing financial mechanisms and modalities, Fund internal acts and other environment protection documents (guidelines, by-laws, etc.) will be developed through UNDP procedures by niche experts (legal, energy, financial). Afterwards, UNDP, together with Fund, will facilitate the enforcement and operationalization of developed financial mechanisms (grant, revolving, loans…) based on previous activities.

- Once financing mechanisms and modalities are developed and operative, end-user's will have more financial options to invest in energy efficiency, while setting an example for residential and commercial buildings, as well as providing the proof to BiH based financial institution sector (government and commercial) that energy efficiency investments are economically and financially attractive and cost-effective.

- Activity 7 sets the basis for all other activities, especially Activity 8, as well as Activity 9 (in order to communicate to the general public the benefits of EE investments and possibilities to finance those investments).

Activity 8 - Implementation of infrastructural energy efficiency measures in public sector buildings (refers to PC4):

- Activity 4 and Activity 3 will provide EMIS and energy audits based evaluation of energy and cost saving potential as well as financial calculations based on which a ranking of public sector buildings will be developed in order to make decision on infrastructural EE investments.

- On the basis of technical, economic and environmental parameters, from detailed energy audits, infrastructure measures will be implemented in up to five public sector buildings per year in order to conduct so called "Deep retrofits" (in accordance with the requirements of EU directives EED and EPBD) and respecting newly developed and adopted construction and energy conservation standards (EPBD directive). The ranking list/evaluation will include the best case techno-economic scenarios developed under Activity 3 (detailed Energy Audits).

- Once the list is developed UNDP will arrange co-financing from end users (third-parties), develop and sign Cost Share Agreements on co-financing the implementation of infrastructural energy efficiency measures.

- The next phase is followed by procurement process for development of design documentation which includes publication of a public tender/call for qualified bidders, evaluation of bids and Contract signing. As per UNDP's Financial Regulations and Rules, the main principles which guide UNDP procurement are: Best Value for Money, Fairness, Integrity, Transparency, Effective International Competition and the Interest of UNDP.

- After contract has been awarded, the contractor will start working on development of design documentation (architectural, construction, mechanical, electro etc.), technical descriptions, drawings, bill of quantities etc. During design process UNDP will hire independent revision in order to ensure quality assurance of designed technical documentation and after that will start bidding process and selection of the deep energy retrofit contractor. Common energy retrofit measures are: improvement of the thermal insulation of the building envelope (insulation of external walls, roofs, windows replacement), improvement the efficiency of building equipment, improvement of the automatic control systems and use renewable energy generation sources (fuel switch from fossil fuels to biomass).The quality of site supervision has a major influence on the overall performance and efficiency of construction projects. Inadequate supervision is believed to be one of the major causes of rework. Therefore, quality assurance will be ensured by UNDP through an independent on-site quality control (works supervision) engineer who will closely monitor all executed works, materials used and absence of the work schedule/plan. This engineer/s will be independently contracted just for this purpose. Additionally, Investors (UNDP) supervision will be present on the filed who will additionally monitor the works and

progress and inform of any discrepancy from the contracted works or plans. Payments will be done upon the works completion and following signing of monthly payment certificates by both construction supervisory entities: Independent company for supervision of works hired by UNDP and UNDP Engineer. Moreover, warranty period for the works performed will be 2 years following the technical acceptance of the works, during which period the contractor will remove all eventual defects relating to their work. The warranty period will also be covered by the Performance Guarantee (10% of the contract value).

- After the project has been designed, built, and commissioned, its energy consumption needs to be measured. This will be achieved by installing calorimeters for energy consumption monitoring (heating, electricity and water) and its integration with EMIS system. This is very important part of the project, since concrete measures are implemented, and possibility to measure before/after indicators is enabled, and gives visible outputs that are easily demonstrated to public. The generated savings will be reinvested into other EE projects as per EEAP and/or through the obligation in respect to financial mechanisms (a description on project sustainability and the obligation to reinvest generated savings into new EE projects/measures, or to use the savings to finance the investment, is further described in Chapter 4.5.).

Activity 9 - Raising public awareness in the field of energy efficiency, energy management and reduction of emissions to air (refers to PC5):

- Awareness raising campaign will be conducted annually, aiming to ensure change of perception, knowledge, attitudes and practice of children age 13+, youth in general as well as general population countrywide (with particular focus in 15 cities countrywide). Campaign aims to achieve positive change by at least 15 percent annually (measured using pre and post campaign surveying of targeted audiences).

- GED project will roll-out annual campaigns by building on positive experience of promotional campaign conducted in 2014, using already established and well received branding (Professor Atom). Selected contractor will each year, together with GED representatives, recommend most appropriate combination of communications tools and channels, to ensure efficient and effective fulfillment of campaign goals.

- GED project will aim to reach no less than 100,000 people through a combination of communications tools and channels (including TV and radio), and no less than 4,500 persons (direct communication at open-air promotional events in 15 cities).

- Use of multiple communications channels and tools will ensure access to above mentioned targeted audiences: TV spots, radio jingles, social networks, open air events and public appearances in all events related to energy efficiency and innovation, media advisories, press releases, op-eds, etc. Selected contractor will be tasked to produce and disseminate all communications materials.

- All participants in open-air promotional events will be motivated to actively contribute to smart energy use (by using the approach where each person will be asked to choose at least two energy efficiency advises he/she will adopt and use at home).

- Each year series of promotional events will be held in 15 cities countrywide as an integral part of the promotional campaign. Open air events will be designed to attract target audiences (children age 13+, youth in general as well as general population living in those cities), inform and motivate them to change their attitudes and practice regarding smart energy use. Brochures and leaflets on energy efficiency (developed by UNDP) will be disseminated at each event to no less than 500 persons. Quiz game on energy use and how to save energy will ensure interaction with the audience at these events.

- Children age 13+ and youth will be engaged by organizing visits of lead campaign character, Professor Atom, to primary and secondary schools at each of the 15 cities, in cooperation with schools' management. Professor Atom will speak about the energy efficiency and draw attention to the fact that all of us contribute to energy efficiency. Selected contractor will be tasked to coordinate all of the mentioned events with designate local authorities and school management.

- Contractor in charge of conducting promotional campaign will also be tasked to conduct survey prior to and upon completion of the promotional campaign, to ensure accurate (qualitative and quantitative) measuring of change in perception and attitudes of targeted audiences. This will be done by using questionnaires and online surveying (the latter will target social network users who will be targeted with the campaign).

- GED project will develop promotional materials that target each of targeted audiences. These include but are not limited to TV spots (animated videos, cartoons), radio jingles, social network posts/adds educational leaflets and brochures that explain basic to advanced principles of energy efficiency, management and emission reduction. EE from the basics to the technical details related to energy savings, energy management and emission reduction. In addition to producing promotional materials for youth and general public, GED will develop specialized promotional materials for professionals working in the energy sector. In addition, the development of an informative web portal on energy efficiency, energy management and reducing air emissions might be conducted.

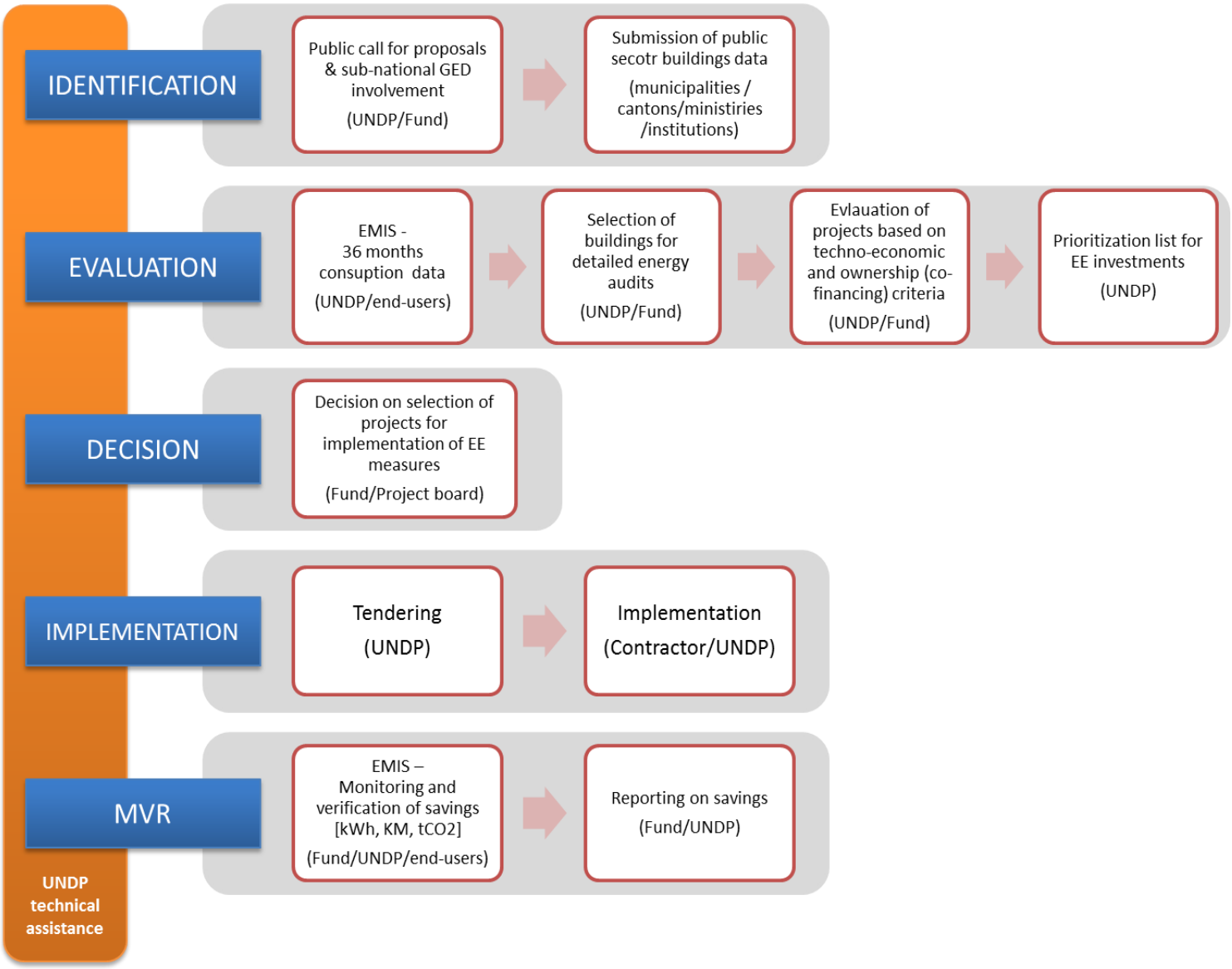
- UNDP will contract the development of: i) short animation movies on EE e.g. production of 20 cartoons and 40 Vine-style videos promoting smart energy use and energy efficiency (in BCS language), ii) Marketing Campaign (media-buying, news reporting, PR assistance, social networks and raising public awareness events BiH cities, development and printing of promotional material on EE), iii) public opinion analysis of target groups prior to and upon completion of awareness raising campaign in order to measure the campaign success and change in attitudes and perception of targeted audiences

Awareness raising campaign and production of promotional materials will provide a follow up on all GED activities and communicate conducted activities during the project. Building the public awareness on energy efficiency will bring changes both in public and private sector.

3.3.4. Methodology on selection of public sector buildings for the implementation of energy efficiency infrastructure measures

Under the GED project activities, the *Methodology on selection of public sector buildings for the implementation of measures to increase energy efficiency* has been developed and its validity justified in 2013 on activities conducted within Environmental Protection Fund of FBiH. Activities conducted during the implementation of the given Methodology contribute to increase internal capabilities, skills and develop capacities within Environmental Protection Funds on strategic approach to energy efficiency investment decision making.

Key Methodology steps are given below, while detailed description is provided in Annex 1.



Methodology - example case from 2013 activities

Based on the methodology, out of a total number of 560 public sector buildings which applied for EE infrastructure financing to Fund in FBiH and UNDP, 40 most promising buildings were equipped with Energy Management Information System (EMIS) software and, based on technical, environmental and economic indicators from the software, 12 have been chosen for further analysis, e.g. detailed energy audits. Based on technical, environmental, economic, financial and social parameters gathered from energy audits, a decision to invest into four EE infrastructure projects has been made (total amount of 420,000 USD, financed by Environmental Protection Fund of FBiH and end-users).

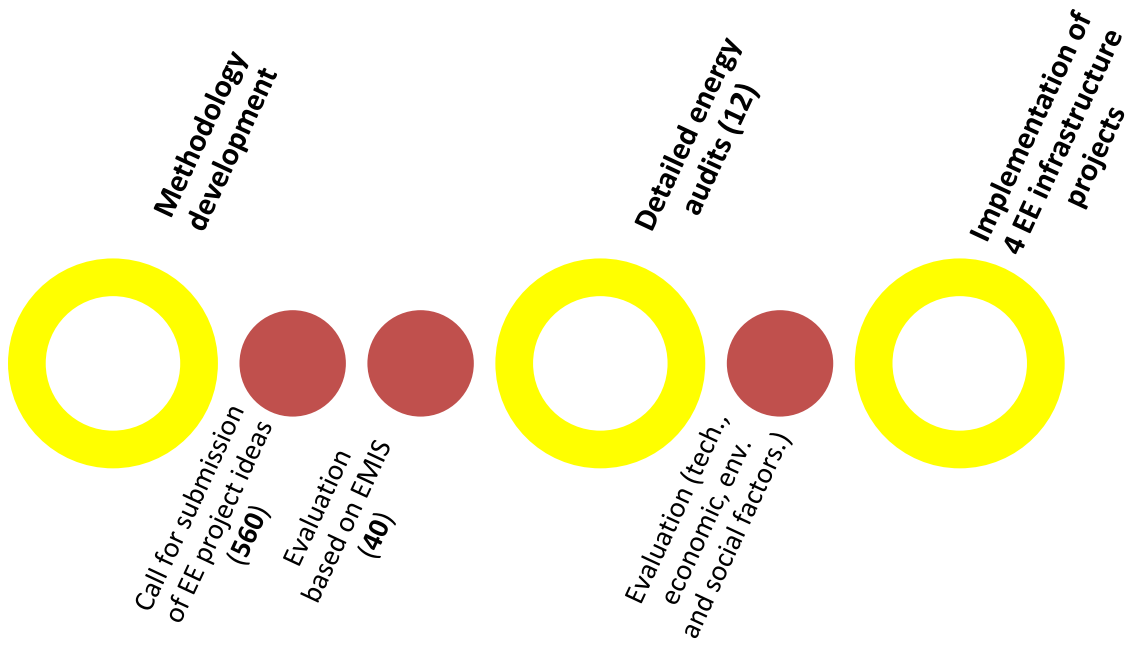


Figure 3 – EE investment decision making process flow

3.3.5. GED project financing

The Green Economic Development project is currently funded or supported by several BiH institutions, while the biggest financial contributor is the Environmental Protection Fund of FBiH (4.2 m USD for the period 2014 – 2018). Institutions funding or supporting the GED project are:

- Environmental Protection Fund of FBiH
- Canton Tuzla
- Canton Una-Sana
- Canton Central-Bosnia
- Canton Western Herzegovina
- Environmental Protection and Energy Efficiency Fund of RS
- Ministry of education and culture of RS

Additional co-financing for infrastructure measures (Project Component 4) is ensured through end-users (municipalities, sub-national governments and institutions), e.g. through a competitive ownership based co-financing ratio evaluation.

3.3.6. Project sustainability and legislative development processes

The aim of GED project activities is to create self-sustainability and ensure that savings generated through implemented energy efficiency infrastructure measures are continuously reinvested into other energy efficiency projects (or other infrastructure projects) or are used to finance the investment.

Sustainability will be ensured by:

a) **creation of energy efficiency revolving fund within Environment Protection Funds**;

b) **sub-national government/end-user obligation to reinvest generated savings and/or to invest into other EE projects** (or other infrastructure projects), based on Energy Efficiency Action Plans (EEAP) and sub-national level based environment protection strategic planning and budgeting; Namely, in accordance with the Law on EE, EEAPs (covering sectors with indicated plans on improvement of energy efficiency) and Operational EEAPs (covering sectors with operational plans, allocated financial resources, targeted public sector buildings and timeframe for assigned actions) will require from end-users to allocate financial resources within their budget (municipal, cantonal and ministerial) for EE infrastructure projects and, simultaneously, to reallocate generated savings to budget lines for additional EE infrastructure projects, and thus enabling them to reinvest savings into new projects. Even without the Law on EE, reinvesting into EE infrastructure from generated savings is possible under the “sanitation/renovation/maintenance” categories within public budgets, with a clear two to three year plan on investments (which includes boiler house/heating system, windows, envelope and other construction works). Therefore, it is suggested that, as an additional monitoring mechanism, institutions/end-users (responsible for energy costs),

sign a Statement on reinvesting generated savings into other infrastructure projects and report back to UNDP on allocation of generated savings.

c) **Law on Energy Efficiency, based on EU's EPBD and EED directives** (which identifies Environmental Funds in both entities as responsible institutions to: i) monitoring energy consumption and savings, ii) supporting EE targets iii) supports the implementation of energy efficiency projects; The Law on EE also incorporates EMIS as one of IT tools to monitor and report on energy consumption);

d) **Selection of public sector buildings/decision making process on investments** - Since not all public sector buildings provide attractive and cost-effective financial results, it is very important that a large pool of potential public sector buildings is identified, analyzed and prioritized based on several technical, economic and environmental factors in order to justify the right investment decision. This is made available through i) Project Component 2 - *Institutionalization of energy management*, which sets the baseline pool/building stock of public sector buildings, ii) the developed Methodology (*Chapter 3.3.3. Methodology on selection of public sector buildings for the implementation of energy efficiency infrastructure measures*), and iii) conducted detailed energy audits (which include detailed techno-economic and financial analysis of proposed EE measures, e.g. required investments, expected energy and costs savings, expected emission reduction, pay-back period, NPV, IRR etc.).

Moreover, currently, and throughout the GED project implementation period, various legislative processes are on-going which are aligned and in favor of GED project's sustainability and creation of financial mechanisms within Environmental Funds.

Repayment of co-financing amounts to the Energy Efficiecny Revolving Fund, as well as the reallocation of energy costs savings to other EE projects/other infrastructure projects, will be assured by:

- Fund is a financial institution, already some financial mechnisms in place
- Law on EE
- Budgeting EE savings and allocation to other EE activites/projects
- Secondary legislation on budgeting energy savings and Law on procurement
- Obligation to invest into EE project as of EEAPs
- Obligation through financial mechanisms of Environmental Protection Funds

3.3.7. Project rights' perspective

A rights perspective has been considered in the project document design, specifically with regard to ensuring that public policies, legislation, financing mechanisms and service outcomes to be supported within the project are non-discriminative and offer equal opportunities (access to services, funding, employment opportunities) for all, regardless of their age, ethnic, sex or social

status. Highly participatory approach in policy-making, legislating or funding assisted by the project reinforces the principle of transparency. Ensuring wide publicity in the course of project work and making publically available policy deliverables and measurable results reinforces the accountability principle.

3.3.8. Gender equality

Gender equality has been mainstreamed within the project implementation strategy. Particular attention will be given to ensuring gender-sensitive design of regulatory and policy frameworks, as well as warranting that both men and women benefit equally from services, funding, employment and capacity development opportunities supported by the project. Women will be encouraged to participate equally in activities led by the project.

3.3.9. EMIS/Monitoring, analysis and evaluation of cost and energy savings and emission reduction

Energy Management Information System (EMIS) is web based application for tracking, analyzing and evaluating energy and water consumption in public sector buildings. Application as such represents necessary tool for energy management. In Croatia, 9.000 public sector buildings are currently monitored via EMIS for Law on EE purposes.

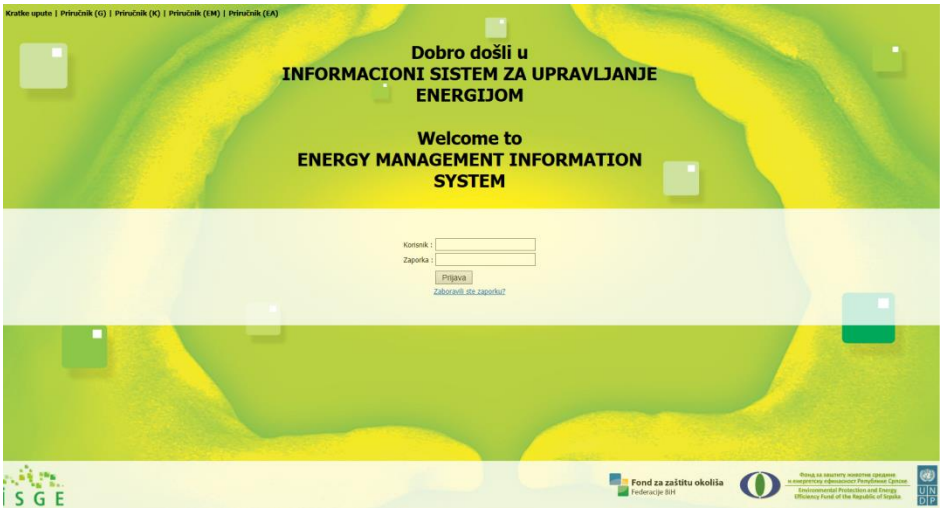
Energy management includes strategic planning and sustainable management of energy resources. For buildings owned by the public sector; municipalities, cantons and entities (administrative buildings, hospitals, schools, etc.), the person responsible for managing energy, input relevant information about buildings that are under their jurisdiction. In compliance with the level of responsibility, each user is assigned certain rights in the software (guest, user, manager, administrator etc.).

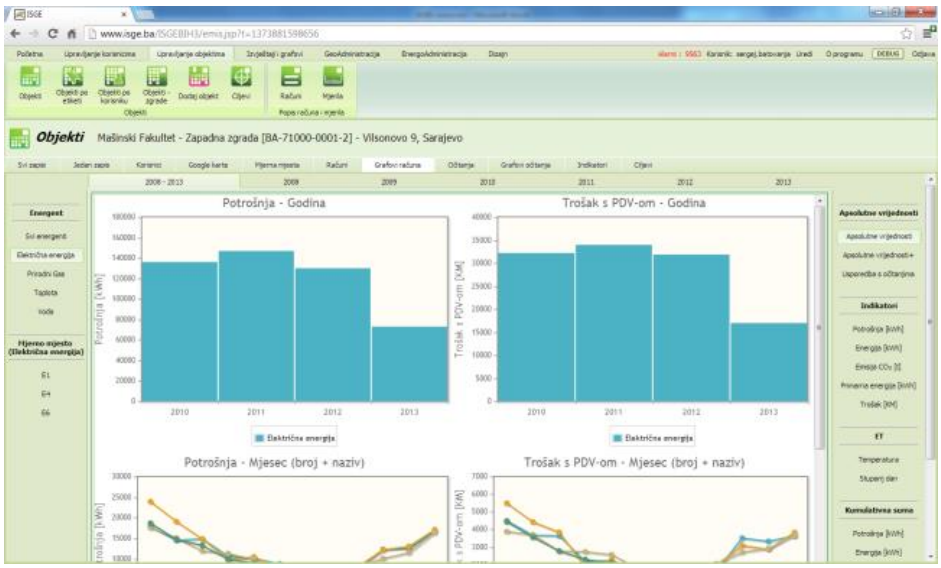
The EMIS users first enter static information about the building, which include construction and energy properties of the building. Then monthly consumption data is entered based on the utility bills that are delivered. Also there is an option to add additional (more frequent) consumption data for each building. This data is either entered manually, by reading it from the meters/energy bills, or automatically by installing automatic remote metering systems, which can provide hourly consumption data directly to EMIS.

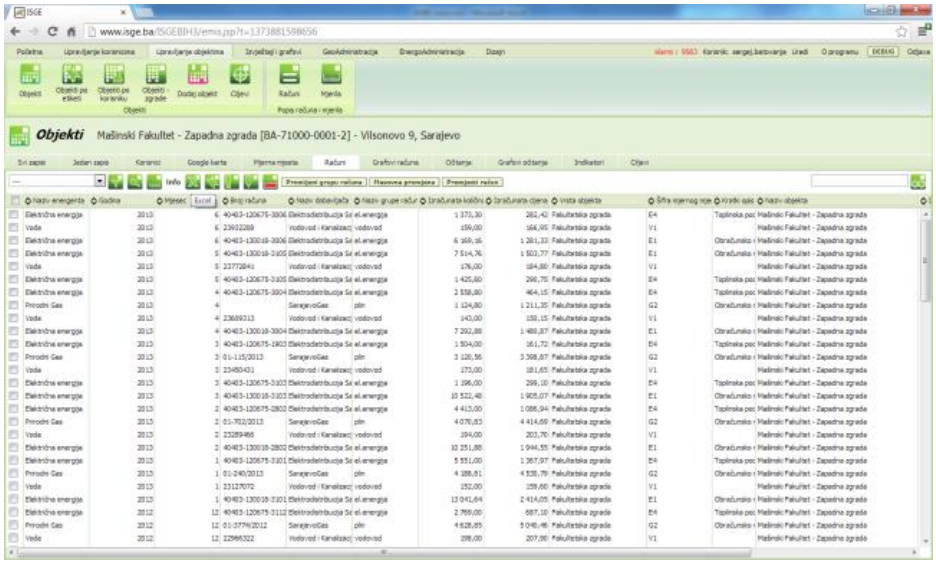
Data entered into system is used for calculating, analyzing and control and enables a better understanding of where, when and how we consume energy and water in buildings. It also allows comparison of spending between the selected buildings as well as the identification of unwanted and excessive spending.

EMIS greatly eases the process of systematic energy management in public sector buildings, in a way which allows easier access to information on consumption and energy costs. It provides a simple graphical representation or a tabular presentation of the data collected as a result of predefined analysis. Continuous monitoring of expenditure through EMIS provides a transparent view and control of energy consumption, energy savings, emission and emission reduction in public sector buildings. Main functionalities of EMIS (more info under www.isge.ba) are:

- Collect and enter basic information about the buildings and the control of energy and water consumption on a monthly, weekly and daily basis
- Easy access to information on energy use, consumption patterns and places of consumption
- Calculations and analysis of consumption for the purpose of identifying unwanted and excessive consumption of energy and water, as well as identifying opportunities to increase energy and financial savings
- Verification of realized energy and water savings (kWh, €, t CO2, and any other standardized indicator)
- Calculation of various consumption indicators
- Automatic notification of accomplished goals or unwanted consumption
- Different options for different levels of users
- Advanced database filtering
- Easy export of data info PDF or Excel formats
- Filtering based on building type


Kratke upute | Priručnik (G) | Priručnik (K) | Priručnik (EM) | Priručnik (EA)
Dobro došli u
INFORMACIONI SISTEM ZA UPRAVLJANJE ENERGIJOM
Welcome to
ENERGY MANAGEMENT INFORMATION SYSTEM
Korisnik :
Zaporka :
Prijava
Zaboravili ste zaporku?
ISGE
Fond za zaštitu okoliša
Federacije BiH


Objekti
Mašinski Fakultet - Zapadna zgrada [BA-71000-0001-2] - Vilsonovo 9, Sarajevo
Potrošnja - Godina
Trošak s PDV-om - Godina
Potrošnja - Mjesec (broj + naziv)
Trošak s PDV-om - Mjesec (broj + naziv)


Objekti
Mašinski Fakultet - Zapadna zgrada [BA-71000-0001-2] - Vilsonovo 9, Sarajevo

IV. SWEDEN'S INVOLVEMENT

In accordance with *Results Strategy for Sweden's reform cooperation with Western Balkans* for the period 2014 – 2020, the Green Economic Development (GED) project is fully aligned to expected result 3 – *A better environment, reduced climate impact and enhanced resilience to environmental impact and climate change*.

EU integration is crucial to Bosnia and Herzegovina's pursuit of democracy, rule of law and sustainable economic development. However, due to its administrative complexity, political lock-down and lack of human and financial resources, Bosnia and Herzegovina needs substantial assistance on its path to EU accession in form of capacity building and strengthening of EU related processes and the transposition of EU directives into national legal framework.

The Green Economic Development project will contribute to BiH's EU accession process, targeting the EPBD (Energy Performance Building Directive) and EED (Energy Efficiency Directive) directives by enabling entity Environmental Protection Funds to monitor, analyze and evaluate energy consumption, costs, emission, energy investments and savings data from public sector buildings, as well as to support BiH's related energy efficiency targets (in accordance with the Energy Community Treaty and its obligations) by providing financial support to EE infrastructure projects and contributing to enable a favorable environment for EE investments in BiH. Moreover, the state of BiH (MoFTER) would be empowered with consolidated, valuable and reliable data for further submission to Energy Community Secretariat and other multilateral agreements (for example – United Nation Framework Convention on Climate Change).

With SWEDEN's involvement in GED project, the promotion of more economical, social and environmental sustainable management of resources in BiH would be enhanced significantly.

4.1. SWEDEN's POTENTIAL INVOLVEMENT

Out of five GED Project Components (PC) SWEDEN's potential involvement in the GED project would most effectively contribute and create add-value on *Capacity building* (PC 1), *Infrastructure EE measures* (PC 4) and *Public awareness* (PC 5) components (market orange in Figure 4).

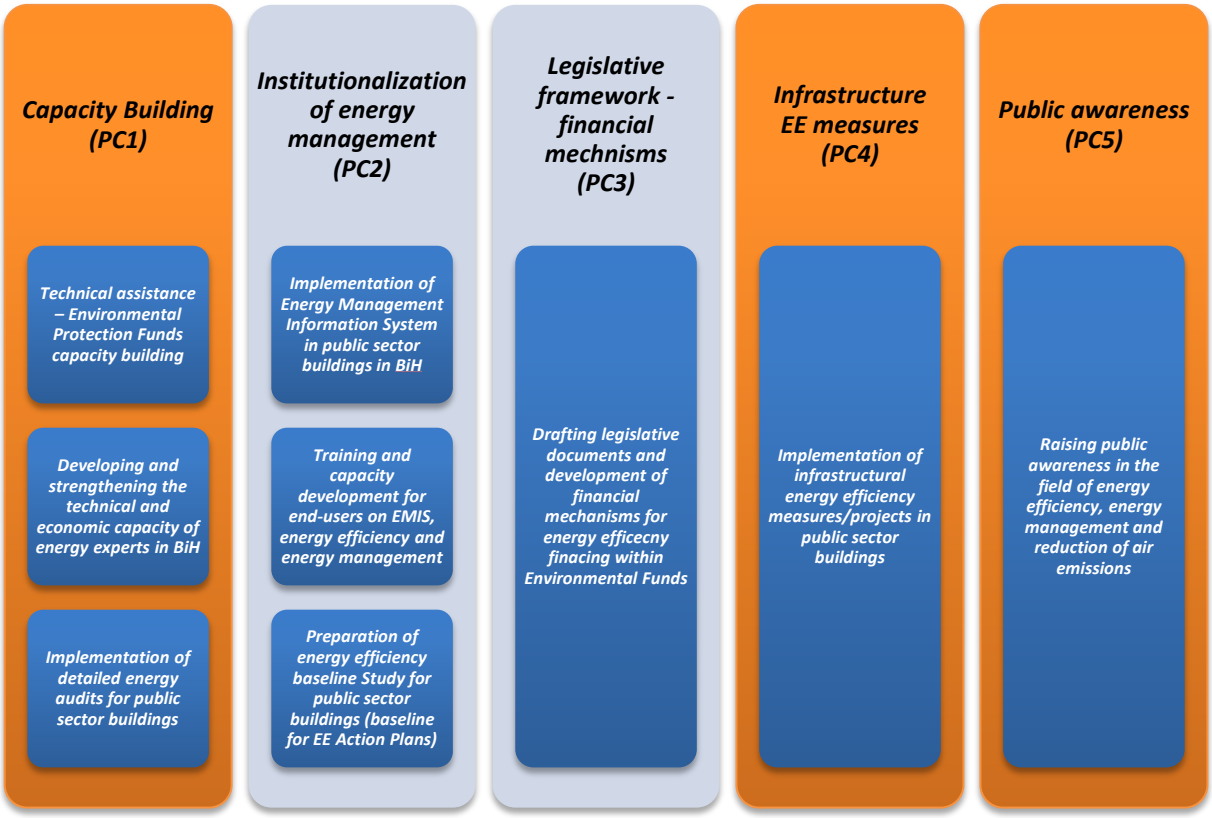


Figure 4 – Overview of identified potential for SWEDEN's involvement

Due to lack of financial resources, only a limited number of energy efficiency infrastructure projects can be implemented under the current portfolio of Green Economic Development project, as well as only a limited volume of Capacity Building and Raising Awareness activities. Moreover, the aim to create an ambient favorable to invest into energy efficiency infrastructure projects, resulting in government financial institutions (Environmental Protection and Energy Efficiency Funds) and commercial financial institutions providing different financial mechanisms to municipalities/end-users of public sector buildings (loans, guarantees, revolving, performance based granting) and the reinvestment of savings (or repayment of loans out of the cost savings achieved), would most probably not be able to occur within the next years. SWEDEN's support would help to increase the number of pilot projects implemented under the current (grant) and to-be-newly-established financial mechanisms (loans), and make them operational.

Therefore, SWEDEN's involvement would significantly contribute to proof BiH based financial institution sector (government and commercial) that energy efficiency investments are economically and financially attractive and cost-effective. Currently, Environmental Protection and Energy Efficiency Funds, as well as commercial financial institutions do not provide loans, guarantees, revolving modalities nor performance based granting for EE measures in public or private sector

buildings. Moreover, there is low awareness on the high technical and economic potential of energy efficiency projects to generate significant cost savings out of which investments could be repaid and further reallocation of savings into other EE projects implemented.

Funds from third party/SWEDEN would also contribute and enable higher volume of activities under Project Components 1 and 3 – resulting in developing and strengthening technical and economic capacity of a larger number of energy experts in BiH and implementation of more energy audits, as well as better execution, coverage and outreach of GED's planned energy efficiency marketing campaign (TV and radio campaign, events organized in 15 BiH cities, publication of raising awareness materials, conferences and events on presentation of GED project milestones and achievements).

More energy audits would significantly contribute to GED project objective, due to:

- higher portfolio of potential projects to be invested in
- enabled better decision making due to better overview of options for potential investments (more public sector buildings to choose from; larger portfolio of technical, economic and financial analysis and therefore better investment planning and decision enabled)
- by increasing financial sources for investments (higher number of public sector buildings which can be potentially retrofitted), the number of energy audits out of which an investment decision should be made shall also increase
- increased possibility to seek for investments from other sources (municipal, cantonal, entity budgets) or financial schemes (once developed in BiH), and therefore leveraging funds invested in energy audits

The civil society / NGO's could also be involved in the execution of the Public Awareness campaign in order to raise further understanding when making EE investments in buildings used for social benefits, as well as to develop and strengthen their own capacities in the area of energy efficiency.

Logical framework and **result and risks matrix**, aligned to SWEDEN's involvement, are provided in *Annex 2* and *Annex 3*.

4.2. ACCELERATING THE CREATION OF A FAVORABLE ENVIRONMENT FOR ENERGY EFFICIECNY INFRASTRUCTURE INVESTMENTS

The intention of GED project activities, among others, is to:

- develop and adopt sustainable financial mechanisms within Environmental Protection Funds for EE project financing (performance based granting, revolving mechanisms, soft-loans, guarantees etc.),
- achieve budget cost savings through implementation of EE projects in public sector buildings,
- reinvest savings into energy efficiency or other infrastructure projects,
- to generate employment of domestic workforce, and
- increase awareness and understanding on EE benefits and proof that energy efficiency investments are economically and financially attractive and cost-effective in BiH.

From an EE financing perspective, SWEDEN's participation in the project would accelerate Environmental Protection and Energy Efficiency Fund's **shifting process from "grant financing of EE projects" to "loans provided to end-users for EE projects"** and "risk minimized/guarantees provided" within the public (and potentially private) sector in BiH, as well as risks mitigated and awareness created for commercial financial institutions (Figure 5).

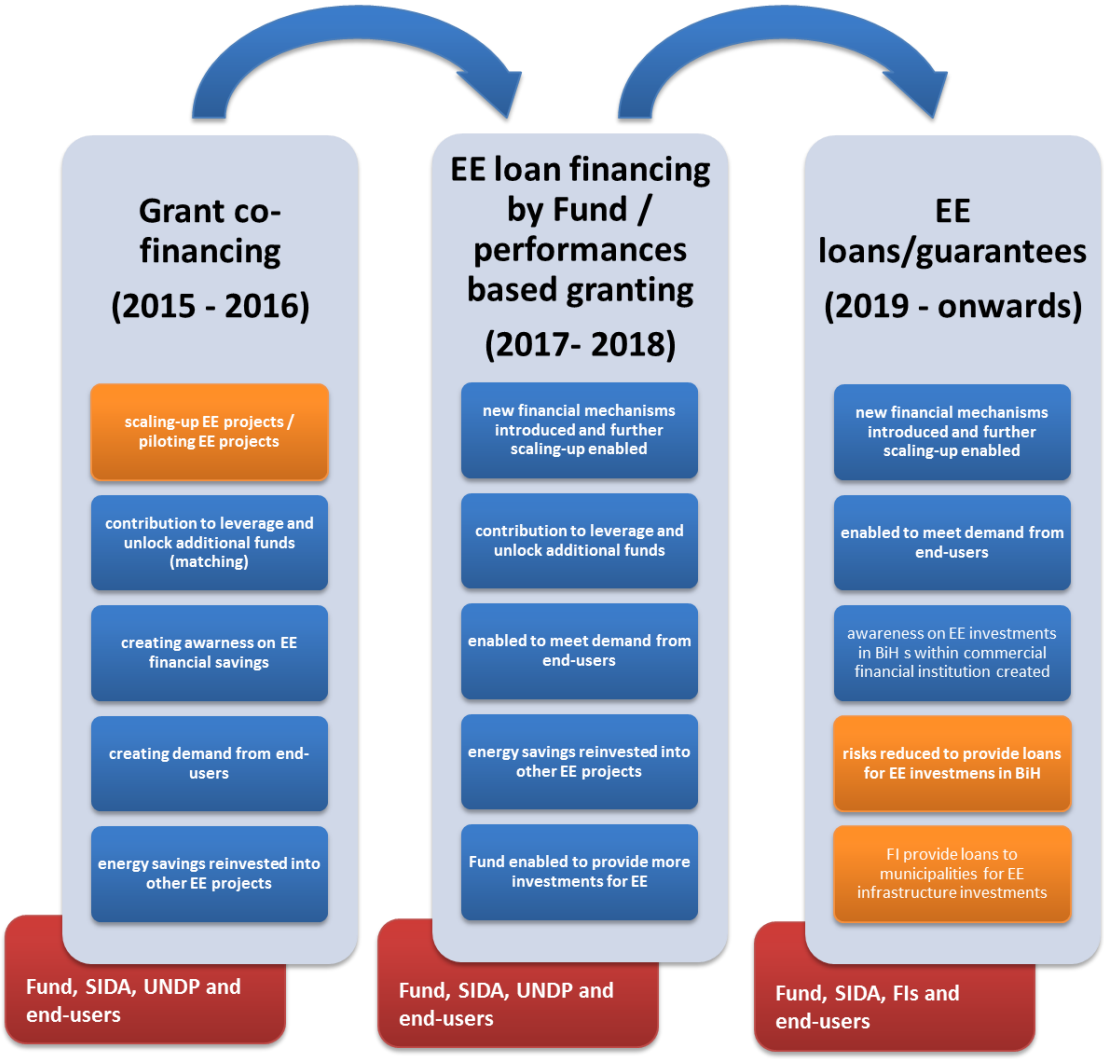


Figure 5 – Financial mechanisms shifting process within Environmental Funds with SWEDEN's involvement

Initial grant financing for infrastructure projects provided by Environmental Funds (currently provided through Environmental Protection Fund of FBiH for public sector buildings in FBiH, while co-financing funds for infrastructure projects in RS are expected to be secured in 2015) and UNDP, do not meet the high interest and demand to unlock additional co-financing possibilities from end users (municipalities, sub-national governments and institutions). A grant financing amount from a third party/SWEDEN supporting the infrastructure component (PC4) would expressively contribute to leverage and unlock additional funds from end-users, while resulting in implementation of a vast number of energy efficiency infrastructure projects and creation of higher ownership rates due to volume of involvement of end-users (Figure 6).

In parallel to those activities, institutionalization of energy management (in order to be able to analyze, evaluate and choose the most promising EE investments/projects, as well as to monitor and verify its savings) activities are conducted aiming to increase the current database of about 1,000 public sector buildings for additional 500 each year in FBiH (on-going) and 500 in RS (as of 2015). Moreover, drafting of legislative framework for financial mechanisms for Environmental Funds in BiH is already being conducted in 2014 (gap analysis, mapping of required legal steps, drafting of

secondary legislation and internal Fund's acts, enforcement of financial mechanisms) and shall continue simultaneously throughout GED implementation period in order to support and strengthen the newly established mechanisms.

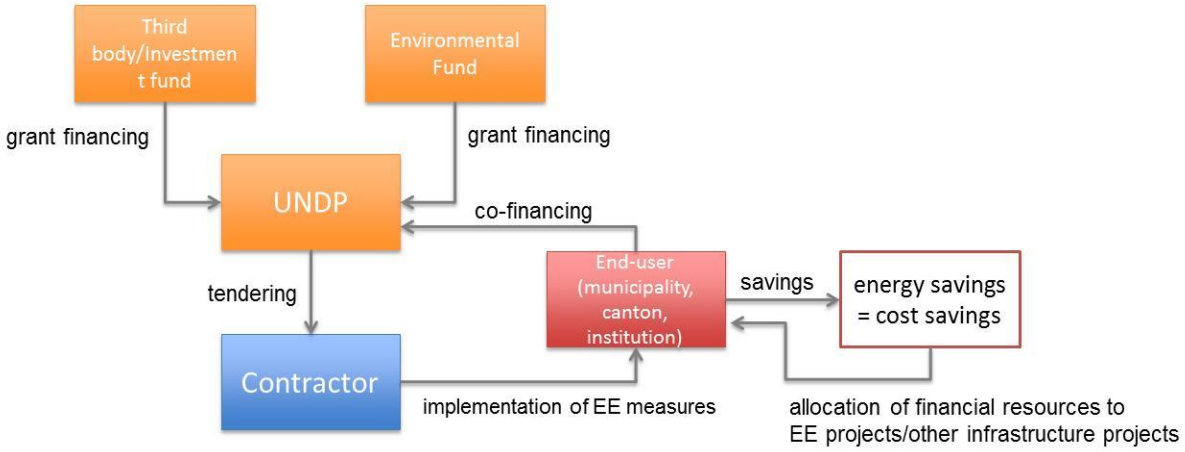


Figure 6 – Grant/matching financial scheme

From 2017 to 2018 it is planned that the Energy Efficiency Revolving mechanism (repayment of generated costs savings to the Fund) within BiH's Environmental protection and Energy Efficiency Funds is enabled, contributing to further scaling-up of EE investments due to higher financial volume supply possibilities from the Fund, and therefore with better abilities to meet the demand from end-users (Figure 7).

Grant financing from a third party/SWEDEN would provide "carrots" in the first phase to motivate end-users to enter into EE investments while further contributing to leverage and unlock additional funds from end-users. Once the revolving mechanisms (and performance based granting) is in place, grant financing from the Environmental Funds could be abolished.

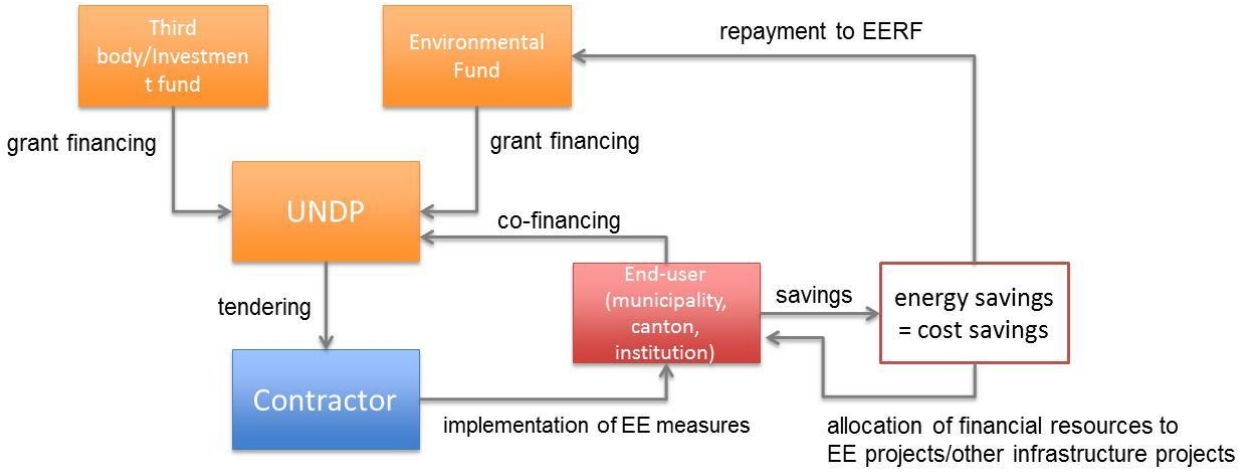


Figure 7 – Revolving financial scheme

From 2019 onwards, it is planned that the Funds provide guarantees and loans for energy efficiency investments in public sector buildings. Moreover, with SWEDEN's involvement in previous two stages, awareness for EE projects and their financial, environmental and economic benefits, as well as motivation and attractiveness to enter into newly developed financial mechanisms would be created within end-users (sub-national level authorities/municipalities/institutions). SWEDEN would significantly contribute to proof BiH based financial institution sector (government and commercial) that energy efficiency investments are economically and financially attractive and cost-effective, resulting in:

- **Risks reduced to provide loans for EE investments in BiH**
- **FIs provide loans to municipalities and other beneficiaries/clients for EE infrastructure investments**

As of 2019, Environmental Protection and Energy Efficiency Funds in BiH would provide performance based granting, revolving mechanisms, soft-loans and guarantees for EE investments in public sector buildings.

Phasing SWEDEN's support

Based on the planned development of Environment Funds financial mechanisms, SWEDEN's support could be phased in two steps as follows:

- Grant co-financing (Environmental Funds and SWEDEN) – 2015 and 2016
- EE Revolving Fund / loan (Environmental Funds) combined with grant co-financing (SWEDEN) – 2017 and 2018

The first phase (in 2015 and 2016) would be grant co-financing of energy efficiency infrastructure projects financed by Environmental Funds, SWEDEN and end-users. All required policy and legal aspects, as well as established mechanisms and modalities for its utilization are currently in place (grant co-financing is already been provided and realized through GED project).

The second phase (in 2017 and 2018) would be the establishment of an EE Revolving Fund within which loans would be provided by Environmental Funds while grants would be provided by SWEDEN (and possibly also Environmental Funds). Once EE Revolving Fund is developed within Environmental Funds, cooperation modalities on the GED project could be further/again defined. Therefore, the key milestone for the second phase would be a fully operational EE Revolving Fund within Environmental Funds, e.g. following steps have been achieved:

- Fund Act contains a basis for the allocation of credits, subsidies and guarantees
- Within internal documents of the Fund, this area is regulated; specified requirements for eligible forms of credits are established
- Plan and program of work of the Fund changed in this regard
- Procedures and mechanisms/modalities for entering into loan contracts developed and enabled

A Study focusing on the further development of energy efficiency based financial mechanisms and modalities (revolving, loans, guarantees etc.), as well as required legislative development within

Funds, is currently been developed. UNDP will support the development of further required internal acts and legislative documents to enable a fully operational EE Revolving Fund within Environmental Funds in BiH.

4.2.1. End-user's affordability

Affordability depends on case by case basis e.g. on end-users' current possibility to co-finance projects or to plan future allocations for projects/investments. It cannot be generalized or assessed on BiH level, nor on entity, regional/cantonal or municipal level. Various types of end-users, depending on the jurisdiction of public sector buildings, have different possibilities to allocate a certain amount of financial sources. Those possibilities depend on:

- General level of governance and motivation
- Budget planning and allocation of resources for infrastructure/maintenance projects
- Level of understanding cost-benefits of energy efficiency investments
- Level of understanding financial mechanisms
- Current energy related conditions of public sector buildings and related expenditures/costs

However, within the GED project it is clearly required and mandatory to have co-financing from end-users in order to be eligible for financing (this is the very first requirement for all public sector buildings in the EMIS pool of buildings, followed by EMIS technical parameter analysis of current conditions, conducting detailed energy audits, conducting techno-economic evaluations, decision on investing, entering agreements, project design, tendering, and finally infrastructure works and commissioning).

SWEDEN's grant co-financing contribution would be co-financed by:

- Environmental Fund
- End-users (institutions, municipalities, sub-national authorities)

Based on current activities and empirical experience on co-financing amounts from end-users, it is expected that SWEDEN's grant financing involvement would unlock additional funds in the matching ratio of 1:1, or leveraging funds of up to two times. Therefore, the maximum co-financing threshold by SWEDEN would be up to 50% (due to the fact that this is a competitive process, the highest co-financing contribution by end-users would be the most competitive from the financing perspective; while technical, economic and environmental parameters are also assessed and included into decision making).

4.2.2. Relation to other EE financing schemes in BiH

This project does not disturb or undermine any other efforts that offer EE finance, but would rather contribute to create awareness on EE financing, reduce risks and create synergies if other new financing sources would emerge in BiH. Moreover, the GED project would set the basis for other schemes in BiH, while contributing to set up a structure and in the long-term, a marked based financial scheme for EE investments.

Currently there are no donors providing finance for infrastructure projects (USAiD's project has been finished in May 2014 and until then UNDP and USAiD coordinated all activities together, while GiZ provides policy related technical assistance). Commercial financial institutions (banks and micro-credit foundations) do not provide financing for the public sector but rather loans without incentives which makes them not attractive.

SWEDEN's involvement would significantly contribute to speed up the development of EE financial mechanisms for public sector buildings while setting an example for residential and commercial buildings (private sector), and would proof to BiH based financial institution (government and commercial) that energy efficiency investments are economically and financially attractive and cost-effective.

4.2.3. Ownership by end-users

Ownership by end-users will be assured through their participation in co-financing EE investments. GED project will provide incentives in form of grant financing and will merge funds from all parties involved, resulting in an attractive modality to invest into EE. However, the goal is to create awareness on benefits of investing into energy efficiency projects, e.g. that savings are used to finance the investments, which makes EE investments interesting for end-users, authorities and general public.

A detailed description on project sustainability closely linked to ownership creation and the obligation and/or necessity to reinvest generated savings into new EE projects/measures, or to use the savings to finance the investment, is described in Chapter 3.3.6.

4.4. EFFECTS OF SWEDEN'S INVOLVEMENT IN GED

With funds secured through SWEDEN's involvement in the amount of 3.6m €, and additional co-financing by end-user's, following effects would be achieved:

- **higher co-financing amount from end-users (matching 1:1, or leveraging up to two times)**
- **about 846,000 € co-financing by end-users/project partners ensured per year**
- **motivating end-users to enter into EE investments (and later into the newly established financial mechanisms) while further contributing to leverage and unlock additional funds from end-users**
- **15 additional infrastructure projects implemented per year**
- **possible higher investment volume and resulting generated savings per project**
- **about 195.000 € additional annual energy costs savings achieved**
- **proof provided to BiH based financial sector (government and commercial) that energy efficiency investments are economically and financially attractive and cost-effective**
- **additional 260 man-months direct employment/"green jobs" generated per year**
- **additional 865 tons of CO_2 emission reduced annually**
- **higher utilization rate of conducted detailed energy audits and implemented EE projects**
- **at least 10 additional detailed energy audits conducted per year**
- **enabling additional energy professionals to participate on capacity buildings trainings**
- **diversified GED coverage throughout BiH**
- **improved execution, coverage and outreach of GED's marketing campaign**
- **scaling-up and creation self-sustainability within energy efficiency related activities in BiH and contribution to BiH's EU accession and fulfilment of Energy Community Treaty obligations**

4.4.1. Expected CO2 emission reduction

Expected CO_2 emission reduction, with and without SWEDEN's involvement in GED project, is given below.

Table 3 – Expected CO2 emission reduction

	2014	2015	2016	2017	2018	Total GED 2014 - 2018
CO2 emission reduction without SWEDEN [t CO_2]	260	260	260	260	260	**1,300**
CO2 emission reduction with SWEDEN's involvement [t CO_2]	260	1,125	1,125	1,125	1,125	**4,760**
Aggregated total [t CO_2]	260	1,385	2,510	3,635	4,760	**4,760**
UNDP's MDGF-Environment project 2010 - 2013, 38 pilot projects [t CO_2]	-	-	-	-	-	2,200

SWEDEN's involvement in GED project would result in enabling additional infrastructure projects to be implemented (15 annually) and thus additional CO_2 emission to be reduced – total 4,760 in the period 2014 - 2018. Namely, SWEDEN would contribute to additional 865 tCO_2 reduced annually, and 3,460 additional tCO_2 during project duration (including 2018, where it is assumed that the EE Revolving Fund will contribute to a greater number of implemented projects, as well as that the understanding of EE investments and readiness to enter EE investments by end-users is higher).

Compared to similar activities in BiH, UNDP's MDGF-Environment project (2010 – 2013) resulted in the implementation of 38 EE pilot projects in BiH and an emission reduction of 2,200 tCO_2.

4.4.2. Immediate action possible

As of July 2014, UNDP and Environmental Protection Fund of FBiH will have 38 detailed energy audits of public sector buildings on disposal - ready for implementation of infrastructure projects. Due to current available funds (200,000 € UNDP/Fund, and up to 200,000 € co-financing from end-users/municipalities/sub-national governments, based on matching principal), it is expected that only up to four projects will be implemented. SWEDEN would contribute to scale up infrastructure projects and generate a far bigger co-financing volume from end-users, resulting in more total energy savings and reinvestments in further energy efficiency infrastructure projects (or other infrastructure projects).

4.4.3. Remarks

- All funds provided by project parties involved in GED (Fund, SWEDEN, end-users/sub-national governments, UNDP) would go through UNDP. UNDP would be responsible for financial and project management assuring a transparent use of funds, clear and effective processes and political neutrality.

- All funds provided by third party/SWEDEN would, on an annual basis, be subject of external financial audit revision.

- Third party/SWEDEN's visibility obligations and guidelines would be fully respected within all GED activities, produced documents and raising awareness marketing campaign.

V. PROJECT ARRANGEMENTS - MANAGEMENT & REPORTING

5.1. MANAGEMENT ARRANGEMENTS

UNDP's role

Management Arrangements of the Green Economic Development project are already in place. UNDP Country Office in Bosnia and Herzegovina is the Implementing Entity for the project and would be accountable to SWEDEN for the disbursement of funds. Working closely with project partners (Environmental Funds), UNDP will be responsible for: (i) providing project assurance services (ii) recruitment of project staff and contracting of consultants and service providers; (iii) overseeing financial expenditures against project budgets; and (iv) ensuring that all activities including procurement and financial services are carried out in strict compliance with UNDP procedures. A UNDP Project Manager is already assigned with the responsibility for day-to-day management and control over project finance.

UNDP Country Office in Bosnia and Herzegovina will provide full support to the national project implementation and will assume full responsibility and accountability for the overall management, including monitoring and evaluation of project interventions, achievement of project outputs and specified results, and the efficient and effective use of resources[12].

The UNDP country office shall provide support services for the Project as follows: (i) HR activities including recruitment of project personnel, issuance of project personnel contracts etc; (ii) process of undertaking procurement processes for acquiring of project goods and services; (iii) financial transactions; etc. The UNDP Country Office in Bosnia and Herzegovina offers the following dedicated staff capacity for project implementation support in the area of environment and energy: (i) Energy and Environment Sector Leader who oversees programme implementation on daily basis, including quality assurance and monitoring and evaluation; (ii) Programme Support Officer - reviews the budgets and monitors project delivery status; (iii) Head of Operations Unit - assures compliance with overall fiduciary standards of UNDP; (iv) UNDP Resident Representative, who shall liaise with high-levels of with the Government and will negotiate key policy changes proposed by the project.

UNDP Bosnia and Herzegovina will maintain the oversight and management of the overall project budget. It will be responsible for monitoring project implementation, timely reporting of the progress to SWEDEN, as well as organizing mandatory and possible complementary reviews and evaluations on an as-needed basis. It will also be responsible for procurement of the required expert services and other project inputs and administer the required contracts. Furthermore, it will support the co-ordination and networking with other related initiatives and institutions in the country.

[12] The roles and responsibilities for the implementation of the project will be in line with the global UNDP User Guide on Program and Project Management for Results which defines minimum requirements for ensuring accountability for programmed activities and use of resources.

Environmental Funds' role

The Environmental Funds' in BiH are project implementation partners whose role is to develop capacities and gradually contribute more and more to project implementation activities as their skills raise during project implementation. Environmental Funds' representatives work on daily basis together with UNDP project staff, where technical assistance in the field of energy efficiency, energy management and management of EMIS (*Energy Management Information System*), understanding of Laws, by-laws and EU directives is provided. Environmental Funds' apsorb the assistance in the form of consultative, advisory and technical support, through annual training for Fund staff and project implementation and skills development via on the job training/assistance on all phases of project (more information provided in chapter 3.3.3. *Description of GED planned activities, Activity 1*).

Environmental Protection Fund of FBiH

The Environmental Protection Fund of FBiH derives its operating income mainly from fees charged to polluters and natural resources users, as well as from fees levied on motor vehicle registration and donor funds from international cooperation. Total income for the Fund amounted to about BAM 60 million (about EUR 31 million) in 2013. The Fund has about 30 employees, and the administrative budget represents 6.7% the Fund's annual income. It has implemented 422 projects over the 2010 – 2013 period.

The Fund requires at least 20% co-financing by the end-user of a project (30% is required if the end-user is a private sector entity). The Fund is managed by a Steering Committee/Management Board and controlled by the Supervisory Board. The day-to-day operations are managed by the Director, who is appointed and by the Management Board with the approval of the Government of FBiH. The Fund is audited by auditors appointed by the Federation.

The Fund is administratively, economically and technically capable of working with energy efficiency and already participates in the GED Project. UNDP has entered into a five year agreement with the Fund to implement the GED Project.

Environmental Protection and Energy Efficiecny Fund of RS

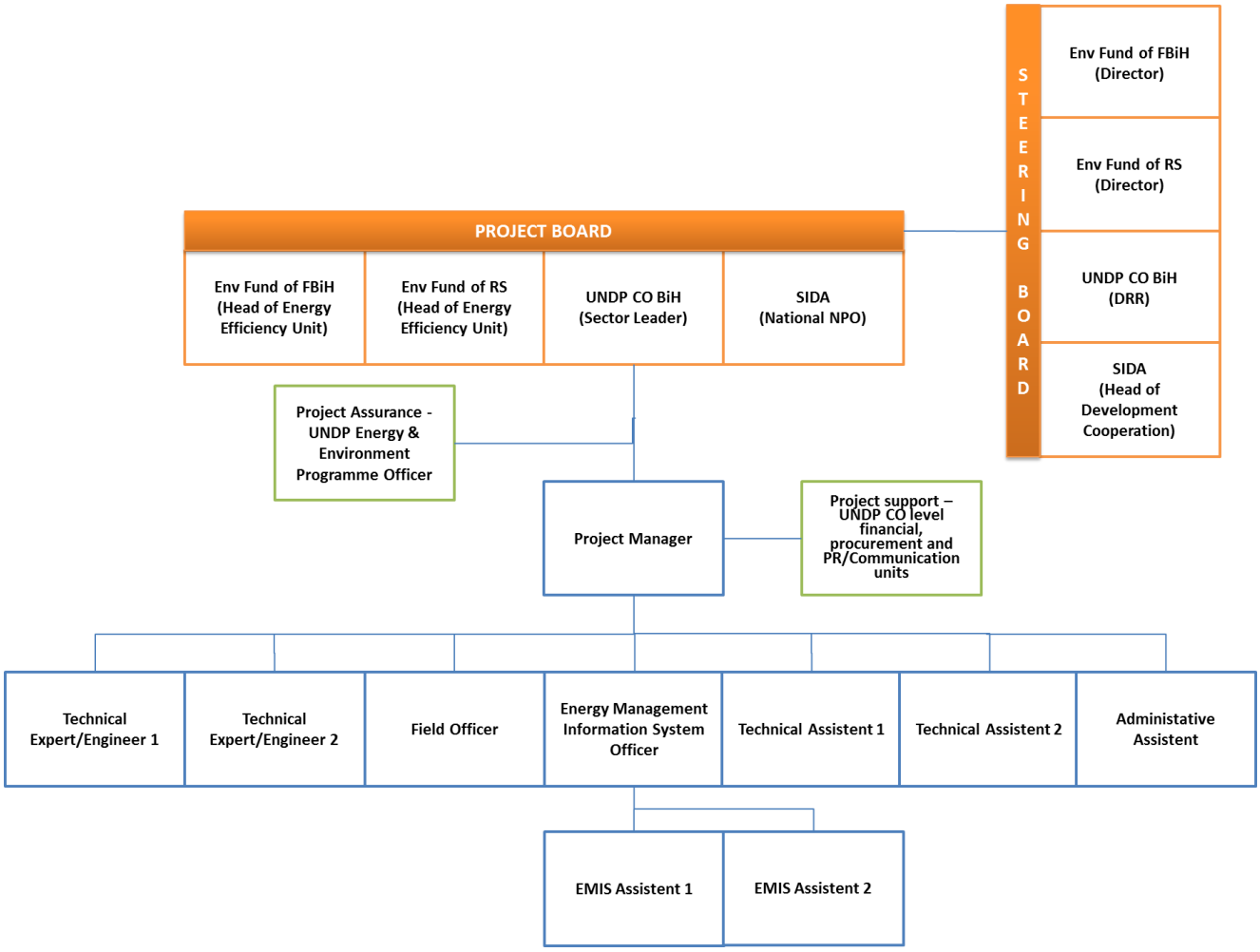
The Environmental Protection and Energy Efficiency Fund of RS currently has only one income source, which is an allocation of 5% of the feed-in tariff that is accorded to energy producers that utilise renewable energy sources. The Fund is also to be allocated 15 % of the proceeds from the water protection fees levied in RS on owners of motor vehicles, once this mechanism is enforced (which is expected beginning of 2015). Other potential sources of income are 'polluter pays' fees and fees for waste disposal as well as donor funds. Total income for the Fund amounted to about BAM 2.4 million (about EUR 1.2 million) in 2013. The Fund has around 30 employees, while the administrative budget is reported to currently represent about 30% of the Fund's annual income.

The Fund requires at least 30% co-financing by the end-user of a project. So far, the Fund has only financed projects related to water quality, mainly in the private sector. The Fund is managed by the Steering Committee, which consists of seven members that are appointed by the Government of RS.

The members are representatives of relevant ministries and authorities. The Director of the Fund is appointed by the Government of RS on the basis of a public competition.

UNDP has signed a Memorandum of Understanding with the Fund in RS concerning the GED Project. Currently, a three to four year Agreement is been prepared. Moreover, participation by the Fund of RS in Project Component 4 (investments in EE infrastructure measures) is likely to be deferred to the time with the Fund will have access to air pollution fees that will give it more financial resources (expected in early 2015). Therefore, an unequal distribution between FBiH and RS of funds for infrastructure measures and energy audits could be expected in the first year of SWEDEN's involvement in GED. However, UNDP CO BiH will give equal opportunities to both Funds and will put additional efforts while supporting the RS Fund in order to be on the same mature stage as the FBiH Fund, and enable them to take full advantage of GED project as well as SWEDEN's involvement. If the Fund of RS should be unable to financially support infrastructure projects, UNDP will make every effort to enable appropriate co-financing from municipalities, ministries and institutions from RS and thus facilitate energy efficiency infrastructure projects in Republika Srpska.

Project Organizational Structure



Steering Board

The Steering Board is planned to be established under the GED project, with SWEDEN involvement in the project. The Steering Board is the highest-level body and as such responsible for the strategic alignment of activities under the GED project with relevant policies of organizations involved, and can act as the body which activities removes barriers, if occurred.

The Steering Board will consist of the following roles:

- Executive role - UNDP Deputy Resident Representative;
- SWEDEN, Head of development cooperation;
- Environment Funds of FBiH and RS, Directors of Environmental Funds.

The Steering Board will meet once a year, or as necessary when raised by the Project Board, to review strategic alignment of project activities, progress and discuss on project plans.

Project Board

The Project Board is the group responsible for making,, management decisions through consensus for a project when guidance is required by the Project Manager, including giving recommendations for UNDP/Implementing Partner approval of project plans and revisions. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance to standards that shall ensure management for development results, best value for money, fairness, integrity, transparency and effective international competition. The Project Board plays a critical role in commissioned project evaluations by quality assurance the evaluation process and products, and using evaluations for performance improvement, accountability and response actions. This group is consulted by the Project Manager for decisions when Project Manager's tolerances (normally in terms of time and budget) have been exceeded (flexibility). Based on the approved UNDP's annual work plan (AWP), the Project Board may review and approve project quarterly plans when required and authorizes any major deviation from these agreed quarterly plans. It is the authority that signs off the completion of each quarterly/annual plan as well as authorizes the start of the next quarterly/annual plan. It ensures that required resources are committed and arbitrates on any conflicts within the project or negotiates a solution to any problems between the projects and external bodies.

The Project Board will consist of the following roles:

- Executive, representing project ownership, and acting as chair of the board, in this case, the UNDP Energy & Environment Sector Leader;
- SWEDEN, National NPO;
- Environment Funds of FBiH and RS, Heads of Energy Efficiency Units.

A Project Board has been established at the inception of the project to monitor project progress (Environmental Fund of FBiH and UNDP), to guide project implementation and to support the project in achieving its listed outputs and outcomes. The Project Board will be enlarged with SWEDEN representative. The Board will meet regularly (at least quaterly), or as necessary when raised by the Project Manager, to review project progress, discuss and agree on project work plans.

Project Assurance

Project Assurance is the responsibility of each Project Board member. However, the Project Board might create detailed instructions how to perform the project assurance. The project assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. Project Assurance has to be independent of the Project Manager. Therefore, the Project Board cannot delegate any of its assurance responsibilities to the Project Manager. The role of Project Assurance will be performed by the UNDP Energy & Environment Programme Officer.

Project Management and project team

The Project Manager has the authority to run the programme on a day-to-day basis on behalf of UNDP BiH. The Project Manager has the responsibility to ensure that the programme produces the

required results that are capable of achieving the benefits defined in this document. He is responsible for day-to-day management and ensures that the project produces the results specified, to the required corporate standards and within the constraints of time and cost. The GED project has one National Project Manager. The project currently employs a core team of one senior technical expert, one EMIS officer, two technical assistants and two EMIS assistants to support the successful project implementation and actions outlined in this document. One additional Technical Expert and one Field Officer will be engaged to support the intensive organization, coordination and delivery of GED project. Project administrative assistance is ensured by one full-time post.

The project personnel is selected on a competitive basis in accordance with the relevant UNDP rules and procedures.

Other expertise and resources

The project deploys expertise in various fields as the need arises, in accordance to GED project activities (further described in Chapter *3.3.3. Description of GED planned activities*). Equipment and vehicles are already in place. Accordingly, no financial resources from SWEDEN will be invested in purchasing of new equipment or assets.

5.2. MONITORING AND EVALUATION

UNDP introduced results-based management as its corporate management approach, so that performance at the level of development goals and outcomes is systematically measured and improved. In that sense, monitoring and evaluation are key factors in helping improve performance and achieve results. Monitoring and evaluation will be performed through the analysis of the results-based quantitative and qualitative indicators outlined in the Project's expected results framework (Annex 3).

The main tools for organising the project monitoring system encompass a series of main documents, providing baseline and target quantitative and qualitative indicators, more particularly:

- Logical Framework (as described within Annex 2 of this project document);
- Result and Risk Matrix (as described within Annex 3 of this project document);
- Annual Work Plans, Quarterly Work Plans, detailed financial annual and quarterly plans (in accordance with UNDP standards).

The project will adopt the following reporting modalities:

- Annual Narrative Report prepared by the project and submitted to SWEDEN;
- Annual Financial Report prepared by the project and submitted to SWEDEN;
- External mid-term project evaluation prepared by independent consultant/s and engaged by SWEDEN;

- Final Project Narrative Report prepared by the project and submitted to SWEDEN;
- Final Project Financial Report prepared by the project and submitted to SWEDEN;
- Final project evaluation prepared by independent consultant/s and submitted to SWEDEN.

Furthermore, Annual Project Reviews will be held late each year and will be initiated by the Project Board and may involve other project stakeholders as required. It shall focus on the extent to which progress is being made towards outputs, and that these remain aligned to appropriate outcomes.

Project evaluation will look at the efficiency, effectiveness, sustainability, and relevance of the project objectives. Such analysis will ensure that quantitative results and achievements based on practical experience are not forgotten, as well as that the project builds upon lessons-learnt, sustains positive results and replicates successful initiatives.

A major mid-term project review will be undertaken in late 2016 in order to provide feedback from stakeholders, and if needed, allow adjustments to the project by the Project Board. The mid-term project review shall also be used to identify and evaluate potential SWEDEN's support within the second project phase (further described under chapter 4.2. - *Phasing SWEDEN's support*).

An independent final project evaluation will be conducted in first quarter 2018, so as to examine the overall project`s effectiveness, efficiency, sustainability, achievements, as well as capture lessons learnt and provide strategic recommendations for consolidation and/or exit-phase of the intervention.

GED project indicators, outputs and outcomes are reflected in the One United Nations Programme / United Nation Development Assistance Framework (2015 – 2019) for Bosnia and Herzegovina, as well as within UNDP's Country Programme Document (2015 – 2019) for Bosnia and Herzegovina.

Annex 1 – DETAILED DESCRIPTION OF METHODOLOGY ON SELECTION OF PUBLIC SECTOR BUILDINGS FOR IMPLEMENTATION OF ENERGY EFFICIENCY INFRASTRUCTURE MEASURES

Methodology development

a) Development of methodology with clear weighing factors for selection (development of prioritization and ranking lists) and decision making on implementation of EE infrastructure measures– preparation of detailed technical and economical evaluation, clearly defined criteria and valuing methodology for identification, evaluation and final selection of public sector buildings (UNDP/Fund)

b) Methodology adoption – (Project Board)

1. STEP – Identification

1.1. **Call for proposals** - announcement of Call for Proposals with incorporated Questionnaire on basic information of end-user's public sector buildings (Fund/UNDP)

1.2. **Submission of proposals by end-users (municipalities, institutions) /sub-national GED partners (cantons, ministries)**

2. STEP – Evaluation

2.1. **Entering basic data of public sector buildings into EMIS database** – based on submitted questionnaire, entering basic characteristics (square meters, square meters heated, building type, energy carrier type, etc.) into database, as well as data on energy and water consumption for the past 36 months (data collected, codified and entered into EMIS by UNDP)

2.2. **Selection of public sector buildings for development of detailed energy audits** – based on developed prioritization list with key energy conservation potential indicators (energy carrier, type of heating/cooling system, net heated square meters, average energy cost [KM/m^2], specific heat consumption kWh/m2, specific emission of tCO2/m2), public sector buildings will be chosen on annual basis for conducting detailed energy audits (UNDP/Fund)

2.3. **Analysis and evaluation of detailed energy audits** – based on energy audits results, e.g. techno-economic, environmental, financial and co-financing criteria, all public sector buildings will be evaluated against each other in order to develop a ranking list of most cost-effective energy conservation investments. (UNDP/Fund)

2.4. **Development of ranking list** – based on detailed energy audits findings, as well as further conducted evaluation of energy, environment, economic and financial

parameters a ranking list of most cost-effective energy conservation investments will be developed. (UNDP/Fund)

3. STEP – Decision

3.1. **Decision on co-financing energy efficiency investments of public sector buildings** – based on completed evaluation and recommended ranking list, decision on co-financing energy efficiency investments of public sector buildings will be made. (Project Board)

4. STEP – Implementation

4.1. **Contracting/Procurement** - proceeding procurement of services and works of selected projects/implementation of EE infrastructure projects (UNDP)

4.2. **Implementation and monitoring** - Implementation of EE infrastructure projects by contractors (including supervision as per Law), investor supervision and QA by UNDP

5. STEP – Monitoring, verification and reporting (MVR)

5.1. **Monitoring and verification of savings and consumption**– clear identification of energy and expenses savings (kWh, KM, tCO_2, specific energy consuption...) through automatic analysis of energy consuption and energy consumption indicators data, processed in EMIS software (UNDP/Fund/end-user)

5.2. **Reporting** – development and drafting of comprehensive reports with information on reduced energy consumption, costs, and emission

Annex 2 – GED LOGICAL FRAMEWORK

LOGICAL FRAMEWORK for GREEN ECONOMIC DEVELOPMENT PROJECT				
	Intervention logic	**Objectively verifiable indicators of achievement**	**Sources and means of verification**	**Assumptions**
Overall goal	The overall objective is to create a favorable environment for investing in EE infrastructure measures in BiH.	Number of infrastructure projects directly and indirectly benefiting from improved of financing mechanisms in BiH for EE investments (target value: 100).	Official project-related reports from BiH authorities at cantonal/entity/state levels. Project reports.	
Outcome / Strategic objective	SO1a: To develop capacity and strengthen skills of Environmental Fund staff SO1b: To develop capacity and strengthen skills of energy professionals SO2: To establish energy monitoring and reporting mechanisms in BiH SO3: To enable financing for EE infrastructure projects in BiH SO4: EE in public buildings scaled up	SO1a: Number of detailed energy audits assessed, prioritized and ranked by Environment Funds (target value: 140). SO1b: Number of energy professionals participating on training programme for skills and knowledge development (target value: 500). SO2: Number of sub-national authorities with formal reporting mechanisms in place (target value: 7 – e.g. 5 cantons in FBiH, 2 ministries in RS).	Project reports and relevant technical/tender documentation. Public media coverage; information posted on websites of local/cantonal/entity/state authorities in Bosnia and Herzegovina.	Environmental Funds have internal capacities and capabilities to manage and continue to operationalize energy efficiency thematic area within the Fund. Environmental Funds continue to finance EE projects offering various financial mechanisms/modalities (performance based grants, loans, guarantees) and thus reduce risks for investors and financial

	in cost-efficient way SO5: To increase general public's understanding of EE benefits	SO3: % of increase of financial resources allocated for EE measures through the Fund by the end of 2018 (target value 100%). SO4: Number of effectively implemented EE infrastructure projects in public sector buildings (target value: at least 85). SO5: Change of general knowledge on EE of wider audience/general public (target value: moderate-to-high general knowledge)		institutions. Energy monitoring and reporting mechanisms are replicated and applied as a harmonized system country-wide. Energy professionals and companies are aware of norms and standards and invest into knowledge and skills development. EE infrastructure measures generate significant number of "green jobs". The general public is aware that EE is an economic/development driver.
Output / Result	Result 1: 140 detailed energy audits of public sector buildings drafted and assessed for EE infrastructure investments. Result 2: EMIS is implemented and continuously updated in 5.000 public sector buildings, EEAPs for public sector buildings are drafted and	1.1. Number of detailed energy audits conducted (target value: 140) 1.2. Number of investment decisions made based on evaluation of detailed energy audits (target value: up to 20 annually) 1.3. Number of Fund staff (FBiH and RS each) participating on annual	Source 1: Project reports and relevant technical documentation (all indicators); Source 2: EMIS – official report by Fund to Government (indicators 2.1., 4.1., 4.2., 4.3., 4.4., 4.5.);	Fund's senior management commitment to enforce energy efficiency thematic area within the Fund. Fund's employees

	adopted Result 3: Financial mechanisms and modalities for EE investments within Environmental Funds are established and operational Result 4: In at least 85 public sector buildings EE infrastructure measures have been abreast implemented with EU required technical standards. Result 5: Public awareness on benefits of energy efficiency, energy management and reduction of emissions to air is increased.	capacity development training on energy efficiency, EMIS and EE policy (target value: 10 annually) 1.4. Number of Fund staff (FBiH and RS each) actively working on EMIS/EE investment decision making process cycle, monitoring, assessing and evaluating energy indicators (target value: 5) 1.5. Number of energy professionals participating on training programme for skills and knowledge development (target value: 100 annually) 1.6. Number of female energy professionals participated on training programme for development (target value: 30 annually) 1.7. Change of in-depth understanding of energy efficiency and renewable techniques and technology, ability to conduct detailed energy audits with "deep-retrofitting" measures (target value: moderate-to-high in-depth understanding and ability) 1.8. Number of members of BiH Association of Energy Professionals (target value: 25)	Source 3 – Energy Community Reports on BiH (indicators 3.1., 3.2., 4.1., 4.2., 4.3., 4.4., 4.5.)	motivated to develop and gain additional skills and apply them in daily routine work. Sub-national level authorities are interested to gain capacity and skills in order to absorb Law on EE obligations and required activities related to energy monitoring and reporting. Energy professionals/ construction companies able to absorb planned annual volume of work. Financial mechanisms and modalities for EE investments are timely been developed within Environmental Funds. Existing end-user's interest to implement and invest/co-finance EE infrastructure projects.

		2.1. Number of public sector buildings monitored through the EMIS database (target value: 5000) 2.2. Number of end-users trained on EMIS (target value: 5000) 2.3. Number of EEAPs developed and adopted by sub-national authorities (target value: 7 – e.g. 5 cantons in FBiH, 2 ministries in RS) 3.1. Number of EE infrastructure projects implemented with grant co-financing modality (target value: 60) 3.2. Number of EE infrastructure projects implemented with loan and performance based granting financing modality (target value: 25) 4.1. Number of implemented EE infrastructure projects in public sector buildings (target value: at least 85) 4.2. % of total energy consumption savings within implemented public sector buildings (target value: >50%) 4.3. Achieved energy class of public sector buildings after EE infrastructure measures (target value: energy class “B”)		Energy professionals/ construction companies are able to perform high quality of work and assure timely infrastructure project implementation.

		4.4. % of total energy cost savings within implemented public sector buildings (target value: up to 20%) 4.5. Amount of direct CO2 emission reduction (target value: 4.760 tCO2) 4.6. Number of generated man-months "green jobs" (target value: 1.465) 4.7. Amount of co-financing infrastructure projects during SIDA's involvement - by end-users/institutions/Environmental Fund (total target by end of 2017: 2,538,000) 4.8. Number of end-users/institutions reinvesting generated savings into other infrastructure projects (target: 85)[13] 5.1. Media campaign outreach achieved (target value: 90000 seconds) 5.2. Number of awareness raising events held in BiH (target value: 75) 5.3. Number of promotional materials distributed (target value: 150.000) 5.4. Number of people reached		

[13] * the indicator „% of increased financial resources allocated for EE investments by end-users/institutions" could be additonally included as indicator if relevant and valid data could be collected during project implementation (UNDP will put efforts in this regard)

		through marketing campaign (target value: 500.000)		
Activities	Activity 1 - Technical assistance to Environmental Protection Funds, EE capacity building and skills development (R1): 1.1. Annual training on energy efficiency, energy management, EMIS, Laws, by-laws and EU directives for Fund staff provided by UNDP staff and/or consultants for niche expertise/knowledge 1.2. Technical assistance throughout project implementation on efficiency, energy management, EMIS relevant EU directives, Laws and by-laws understanding provided by UNDP staff and/or consultants for niche expertise/knowledge 1.3. Skills development via on the job training/assistance provided by UNDP staff and/or consultants for	***Means:*** Activity 1 Technical assistance human resources and staffing (interlinked with all other activities) Engagement of one additional engineer (interlinked with all other activities) Staffing (Project Manager, Chief Technical Advisor for EE, Project Assistant, Project Associate x2, EMIS Technical officer) (interlinked with all other activities)		Strong motivation by the Management and operational staff of the Fund to participate, engage and develop EE window/thematic area of work within Fund. Existing end-users interest for increasing EE within public sector buildings. Co-financing supply side (SWEDEN/Fund/UNDP) increased – co-financing volume for infrastructure projects is significantly higher. Law on EE is adopted/in process of adoption in both entities.

	niche expertise/knowledge on: • Understanding of energy conservation potential • Analysis, evaluation and development of energy indicators and reports out of EMIS database • Analysis, evaluation and development of prioritization list of public sector buildings for energy audits • Analysis, evaluation and development of technical, economic, financial, environmental and social parameters/factors (including co-financing from end-users) and development of ranking list of public sector buildings for the implementation of EE infrastructure measures • Monitoring and verification of energy and costs savings, emission reduction • Tracking of indicators • decision making process cycle/methodology for EE infrastructure investments			Institutional willingness of sub-national level authorities (cantons, ministries) to establish energy monitoring and reporting mechanisms. Governments/authorities recognize EE improvements as a potential development generator.

	(identification, evaluation, decision, implementation and monitoring/reporting) <u>Activity 2</u> - Developing and strengthening the technical and economic capacity of energy experts in BiH (R1): 2.1. Development of ToR for development and annual delivery of training programme for development of energy professionals on: i. understanding of legislative framework in FBiH/RS/BiH on EU directives in the field of energy efficiency and energy management; ii. techno-economic, financial and environmental skills/knowledge development necessary to perform detailed energy audits; and iii. expand knowledge on new technologies, techniques and standards in the area of mechanical, civil, electrical and architectural engineering;	<u>Activity 2</u> Development and delivery of training programme (by consultant) Facilitation and support to establishment of Association (by consultant)		

	2.2. Development of ToR for facilitation and support to the establishment of the Association of Energy Professionals in BiH, in accordance with UNDP technical requirements and procedures (SOPs). <u>Activity 3</u> - Drafting of 140 detailed energy audits for public sector buildings (R1): 3.1. Based on technical, economic and environmental parameters and energy conservation potential from EMIS database (Activity 4), development of prioritization list of public sector buildings for the conductance of detailed energy audits 3.2. Development of ToR for tendering of detailed energy audits, in accordance with UNDP technical requirements and procedures (SOPs). 3.3. Coordination and management of development of detailed energy audits and Quality assurance 3.4. Evaluation of technical,	<u>Activity 3</u> Development of 40 energy audits (by consultant)		

	economic, financial, environmental and social parameters/factors (including co-financing from end-users) and development of ranking list of public sector buildings for the implementation of EE infrastructure measures 3.5. Decision on infrastructure investments up to 20 public sector buildings per year <u>Activity 4</u> - Implementation of Energy Management Information System in 5.000 public sector buildings in BiH (R2): 4.1. Annual open Call for Proposals by Fund and UNDP for identification of public sector buildings/end-users interested for EE investments 4.2. Submission of public sector buildings application to Fund/UNDP by filling out pre-defined survey of interest with basic building data and co-financing potential (sub-national project partners submission of list of public sector buildings in their jurisdiction)	<u>Activity 4</u> Activity 4 financing is provided by the Environmental Protection Fund of FBiH and several Cantons (Tuzla, Una-Sana, Central Bosnia, Western-Herzegovina) and Ministry of Education of RS and is part of several other agreements.		

	4.3. EMIS database update (by UNDP/Fund): • Opening an account (static input data) for the identified public facilities • Collecting information on consumption and costs of energy and water in a period of the last 36 months, adjusted for entry into EMIS • Technical support and assistance to contact persons in identified public facilities • Creation of dynamic data for identified public buildings 4.4. Preparation of technical, economic and environmental parameters and energy conservation potential indicators within EMIS database (by UNDP/Fund for Activity 3) 4.5. Management, administration, maintenance and further development of EMIS (by UNDP) 4.6. Monitoring, error identification and correction activities on database			

	(by UNDP/Fund) 4.6. Monitoring and reporting of achieved energy and cost savings (kWh, KM), CO_2 emission (t CO2) reduction and other specific indicators (by UNDP/Fund) <u>Activity 5</u> - Training and capacity development for end-users on EMIS, energy efficiency and energy management by UNDP (R2): 5.1. Development and delivery of EMIS, energy efficiency and energy management training module for up to 1.000 persons annually (two persons per end-user): • Contact all identified end-users (based on Activity 4.2.) • Organize training facilities throughout BiH, equipment (laptops) and travel arrangements • Conduct training on EMIS (based on Activity 4.3.) 5.2. Technical support and assistance to contact persons in identified	<u>Activity 5</u> Activity 5 financing is provided by the Environmental Protection Fund of FBiH and several Cantons (Tuzla, Una-Sana, Central Bosnia, Western-Herzegovina) and Ministry of Education of RS and is part of several other agreements.		

	public facilities throughout project implementation Activity 6 - Preparation of energy efficiency baseline Study for public sector buildings (R2) and EEAPs: 6.1. Development of ToR for the development of energy efficiency baseline Study for public sector buildings on annual basis (based on Activity 4.2.), in accordance with UNDP technical requirements and procedures (SOPs). 6.2. Coordination and management of development of Study and Quality assurance 6.3. Development of ToR for the assistance to development of Energy Efficiency Action Plans and Reporting Mechanisms for sub-national governments, in accordance with UNDP technical requirements and procedures (SOPs). 6.4. Coordination and management of development of EEAP/RM and	Activity 6 Consultancy services/contractors		

	Quality assurance 6.5. EEAP/RM adoption by sub-national governments (by UNDP) <u>Activity 7</u> - Drafting legislative documents, development of financial mechanisms for energy efficiency financing within Environmental Funds (R3): 7.1. Development of ToR for development of Study on improving and developing of energy efficiency based financial mechanisms and modalities (grant, revolving, loans, guarantees etc.), in accordance with UNDP technical requirements and procedures (SOPs): • public call for proposals on conductance of detailed energy audits • evaluation of offers • entering into contractual agreements with contractors 7.2. Technical assistance by UNDP and development of ToR for	<u>Activity 7</u> Consultancy services/contractors		

	development and legislative enabling of internal energy efficiency acts and documents of the Fund (in accordance to findings from Activity 7.1.), e.g.: • financing modalities • Plan and program of work • Development and legislative enabling of procedures and mechanisms/modalities for entering into loan contracts 7.3. Enforcement and operationalization of developed financial mechanisms (grant, revolving, loans… based on Activity 7.2.) (by Fund/UNDP) 7.4. Development of ToR for development of other documents (guidelines, by-laws, etc.) on environment protection, identified during project implementation and aligned to GED project goals <u>Activity 8</u> - Implementation of infrastructural energy efficiency measures in public sector buildings (R4):	<u>Activity 8</u> Project design (by consultant)		

	8.1. Selecting appropriate facilities for deep retrofits and set performance objectives. • Based on technical, economic and environmental parameters and energy conservation potential from conducted detailed energy audits (Activity 3), development of prioritization list of public sector buildings for the implementation of deep retrofit energy efficiency measures (focused on the EPBD Directive). • The prioritization list of public sector buildings includes the best case scenario for deep savings in building energy consumption, annual energy and cost savings, investment costs, payback period and other relevant indicators. 8.2. Arranging co-financing by end users: • Development of Agreement	EE infrastructure measure (by consultant) Supervision (by consultant)		

	on co-financing the implementation of infrastructural energy efficiency measures, • Signature and entry into force of the Agreement on co-finance measures to increase energy efficiency of end-users. 8.3. Procurement process for development of design documentation • Development of Terms of Reference for the mechanical, electrical and structural part of the building • Publication of a public tender / call for qualified bidders • Evaluation of bids • Contracting / signing 8.4. Design process • Development of design documentation (architectural, construction, mechanical, electro etc.): technical descriptions,			

	drawings, bill of quantities… • Issuance of required permits and elaborates • Independent design document revision 8.5. Procurement process for selection of construction Contractor • Development of Terms of Reference • Publication of a public tender / call for qualified bidders • Evaluation of bids • Contracting / signing 8.6. Construction process • Organizing the construction site • Start up • Construction supervision (Quality assurance will be ensured through an independent on-site quality control – supervision engineer, and additionally investor-UNDP supervision) • Commissioning 8.7. Measurement and Verification			

	(M&V). After the project has been designed, built, and commissioned, its energy consumption needs to be measured. This is achieved by installation of meters for monitoring the energy consumption (heating, electricity, water) and integration with EMIS system. <u>Activity 9</u> - Raising public awareness in the field of energy efficiency, energy management and reduction of emissions to air (R5): 9.1. Development of ToR for development of short animation movies on EE e.g. production of 60 cartoons and 50 Vine-style videos promoting smart energy use and energy efficiency (in BHS language), in accordance with UNDP technical requirements and procedures (SOPs).	<u>Activity 9</u> Animation production (by consultant) EE Marketing campaign (by consultant) Designing and printing of promotional material (by consultant)		

	9.2. Development of ToR for development and delivery of Marketing Campaign (media-buying, news reporting, PR assistance, social networks and raising public awareness events BiH cities), in accordance with UNDP technical requirements and procedures (SOPs). 9.3. Development of ToR for development and printing of promotional material on EE, in accordance with UNDP technical requirements and procedures (SOPs).			

Energy and Environment

Annex 3 – RESULTS AND RISK MATRIX

Outcome objective 1	**To develop capacity and strengthen skills of Environmental Fund staff and energy professionals**											
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**	**Risks and Assumptions**	**Likelyhood**	**What should be done for risk reduction**
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**			
Indicator	Number of detailed energy audits assessed, prioritized and ranked by Environment Funds	0	140	Project reports and relevant technical/tender documentation	30	40	40	30	0			
Indicator	Number of energy professionals participating on training programme for skills and knowledge development	0	500	Project reports and relevant technical/tender documentation	100	100	100	100	100	**LOW** • Lack of Fund's senior management commitment to develop, empower and enforce energy efficiency thematic area within the Fund • Lack of motivation by Fund's employees to develop and gain additional skills • Lack of motivation by energy professionals to develop skills and knowledge	**MIDDLE** • The Fund's current senior management as well as employees are motivated to enter a new thematic area and provide financing support for EE projects • It is the Fund's obligation, in accordance to its statute and by Law on EE to provide support to EE projects • Energy professionals are generally motivated to gather skills and enter new areas of work in order to be able to supply the market with consultancy in EE	• Create a vital commitment, knowledge and understanding of importance of developing Fund's capacities in the thematic area of EE within Environmental Fund • Disseminate to BiH's private and academia sector information on knowlede and skills development reqired to enter new market niches within the EE area of work.
Output objective 1	**140 detailed energy audits of public sector buildings drafted and assessed for EE infrastructure investments.**											
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**			
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**			
1.1.Indicator	Number of detailed energy audits conducted	0	140	Project reports	30	40	40	30	0			
1.2.Indicator	Number of investment decisions made based on evaluation of detailed energy audits (target value: up to 20 annually)	0	85		5	20	20	20	20			
1.3.Indicator	Number of Fund staff (FBiH and RS each) participating on annual capacity development training on energy efficiency, EMIS and EE policy	0	10		10	10	10	10	10			
1.4.Indicator	Number of Fund staff (FBiH and RS each) actively working on EMIS/EE investment decision making process cycle, monitoring, assessing and evaluating energy indicators	0	5		2	2	4	4	5			

1.5.Indicator	Number of energy professionals participated on training programme for skills and knowledge development	0	500		100	100	100	100	100
1.6.Indicator	Number of female energy professionals participated on training programme for skills and knowledge development	0	150		30	30	30	30	30
1.7.Indicator	Change of in-depth understanding of energy efficiency and renewable techniques and technology, ability to conduct detailed energy audits with "deep-retrofitting" measures – measured using baseline and post CB questionnaires (qualitative)	low-to-medium in-depth understanding and ability	moderate-to-high in-depth understanding and ability		low-to-medium in-depth understanding and ability	medium in-depth understanding and ability	medium in-depth understanding and ability	medium-to-high in-depth understanding and ability	medium-to-high in-depth understanding and ability
1.8.Indicator	Number of members of BiH Association of Energy Professionals	0	25		0	10	15	20	25

Outcome objective 2	**To establish energy monitoring and reporting mechanisms in BiH**										
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**	**Risks and Assumptions**	**Likelyhood**	**What should be done for risk reduction**
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**			
Indicator	Number of sub-national authorities with reporting mechanisms in place (target value: 7 – e.g. 5 cantons in FBiH, 2 ministries in RS).	0	7	Project reports and Official Gazette	0	2	4	7	7	**LOW/MIDDLE** • Lack of commitment by sub-national level authorities (cantons, ministries) to establish energy monitoring and reporting mechanisms • Lack of capacity and skills by sub-national level authorities (cantons, ministries) to absorb obligations and required activities	**LOW** • It is of interest and direct benefit of sub-national level authorities (cantons, ministries) to participate in the development and establishment of monitoring and reporting mechanisms due to legislative obligation and resulting financial support due to low affordability to conduct EE investments without co-financing • Low awareness and understanding of sub-national level authorities (cantons, ministries) to undertake required activities	• Continues dialog, awareness raising and capacity development activities with sub-national level authorities • Further extension of GED project to other cantons/ministries

Output objective 2	**EMIS is implemented and continuously updated in 5.000 public sector buildings, EEAPs for public sector buildings are drafted and adopted**								
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**
2.1.Indicator	Number of public sector buildings monitored through EMIS database	1000	5000	Project reports, EMIS – official report by Fund to Government	1000	2000	3000	4000	5000
2.2.Indicator	Number of end-users trained on EMIS	1000	5000	Project reports	1000	2000	3000	4000	5000

2.3.Indicator	Number of EEAPs developed and adopted by sub-national authorities (target value: 7 – e.g. 5 cantons in FBiH, 2 ministries in RS)	0	7	Project reports and Official Gazette	0	2	4	5	7	

Outcome objective 3	To enable financing for EE infrastructure projects in BiH											
	Indicator	Baseline	Target	Source of verification	Results	Results	Results	Results	Results	Risks and Assumptions	Likelyhood	What should be done for risk reduction
		2014	2018		2014	2015	2016	2017	2018			
Indicator	% of increase of financial resources allocated for EE measures through the Fund by the end of 2018	0	100%	Project reports and relevant technical documentation, Official Reports by Funds to Governments	0	25%	50%	75%	100%	**LOW** • Lack of Fund's senior management commitment to develop, and enforce financial mechanisms supporting the EE thematic area within the Fund • Lack of understanding by Fund's employees for introduction of financial mechanisms supporting the EE thematic area within the Fund • Financial mechanisms and modalities for EE investments within the Fund are not timely been established during the GED project indicated timeframe	**MIDDLE/HIGH** • The Fund's current senior management as well as employees are motivated to develop and enforce financial mechanisms supporting the EE thematic area within the Fund • Before any new financial mechanism is introduced within the Fund, UNDP and consultants build capacity among Fund's employees • The development of grant co-financing and revolving/loans with performance based granting has started timely	• Timely, and aligned to Fund's legal possibilities, development and enforcement of financial mechanisms for Environmental Funds • Grant co-financing modalities already in place which assures the basis that most of the GED project indicators could be meet
Output objective 3	**Financial mechanisms and modalities for EE investments within Environmental Funds are established and operational**											
	Indicator	Baseline	Target	Source of verification	Results	Results	Results	Results	Results			
		2014	2018		2014	2015	2016	2017	2018			
3.1.Indicator	Number of EE infrastructure projects implemented with grant co-financing modality	0	60	Project reports and relevant technical documentation, Energy Community Reports on BiH	5	20	20	10	5			
3.2.Indicator	Number of EE infrastructure projects implemented with loan and performance based granting financing modality	0	25	Project reports and relevant technical documentation, Energy Community Reports on BiH	0	0	0	10	15			

Outcome objective 4	**EE in public buildings scaled up in cost-efficient way**											
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**	**Risks and**	**Likelyhood**	**What should be done**
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**	**Assumptions**		**for risk reduction**
Indicator	Number of effectively implemented EE infrastructure projects in public sector buildings	0	85	Project reports and relevant technical documentation, Energy Community Reports on BiH, Official Reports by Funds to Governments	5	20	20	20	20	**LOW** • Financing mechanisms supporting EE project investments not in place • Lack of co-financing possibilities by Fund/UNDP • Lack of co-financing possibilities by end-users • Lack of end-user's interest to invest into EE measures • Lack of end-user's interest to reinvest generated saving into other infrastructure projects	**MIDDLE/HIGH** • Grant co-financing modalities already in place which assures the basis that most of the GED project indicators might be meet • Fund/UNDP have only limited amount of financial sources to support EE infrastructure projects; SWEDEN's involvement significantly scales up the possibility to supply the demand side with co-financing and thus implementation of EE infrastructure projects • Empirical and practical evidence shows that public sector building's end-users have the possibilities to co-finance EE projects to a certain extend • Empirical and practical evidence shows that public sector building's end-users have increased interest to invest into to EE projects (high energy costs, expected increase of energy prices, upcoming obligation to invest into EE as per EE Action Plans, redirection of saving to EE/infrastructure measures) • Law on EE and the defined EEAPs/Operational EEAPs will provide the end-user and institutions in charge of energy costs with requirements and enable them to reallocate the generated savings to other EE measures;	• Grant co-financing modalities already in place which assures the basis that most of the GED project indicators could be meet • Involvement of third party/SWEDEN into GED project in order to assure supply side co-financing • Continues dialog, awareness raising and understanding creation on EE benefits with sub-national/local level authorities • Energy efficiency action plans will oblige sub-national authorities to invest info EE measures within public sector buildings • Enforcement of Law on EE and Operational EEAPs; End-users/institutions are required to sign Statement on reinvesting generated savings into other infrastructure projects and report back to UNDP on activities in this regard (UNDP to monitor and follow up in this regard)
Output objective 4	**In at least 85 public sector buildings EE infrastructure measures have been implemented with EU required technical standards.**											
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**			
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**			
4.1.Indicator	Number of implemented EE infrastructure projects in public sector buildings	0	85	Project reports and relevant technical documentation, Energy Community Reports on BiH, Official Reports by Funds to Governments	5	20	20	20	20			
4.2.Indicator	% of total energy consumption savings within implemented public sector buildings	0%	>50%		<50%	>50%	>50%	>50%	>50%			
4.3.Indicator	Achieved energy class of public sector buildings after EE infrastructure measures	Energy class "E" (average consumption of 220 kWh/m2 a)	Energy class "B" (average consumption <95 kWh/m2 a)		Energy class "C" (average consumption <135 kWh/m2 a)	Energy class "B" (average consumption <95 kWh/m2 a)	Energy class "B" (average consumption <95 kWh/m2 a)	Energy class "B" (average consumption <95 kWh/m2 a)	Energy class "B" (average consumption <95 kWh/m2 a)			
4.4.Indicator	% of total energy cost savings within implemented public sector buildings	0%	20%	Referring to indicator 4.8. Statement by end-user on reinvesting generated savings into other infrastructure projects, Operational EEAPs, Official Reports by relevant institutions	0%	20%	20%	20%	20%			
4.5.Indicator	Amount of direct CO2 emission reduction	0	4760		260	1125	1125	1125	1125			
4.6.Indicator	Number of generated man-months "green jobs"	0	1465		85	345	345	345	345			
4.7.Indicator	Amount of co-financing infrastructure projects during SIDA's involvement (by end-users/institutions/Environmental Fund)	0	2,538,000		0	846,000	846,000	846,000	0			
4.8.Indicator	Number of end-users/institutions reinvesting generated savings into other infrastructure projects	0	85		5	20	20	20	20			

Outcome objective 5	To increase general public's understanding of EE benefits											
	Indicator	Baseline	Target	Source of verification	Results	Results	Results	Results	Results	Risks and Assumptions	Likelyhood	What should be done for risk reduction
		2014	2018		2014	2015	2016	2017	2018			
Indicator	Change of attitudes and perception on EE of targeted audiences (children age 13+, youth and general public)in targeted 15 cities countrywide– measured using pre and post campaign survey, using questionnaires and online surveying tools (qualitative and qualitative)	low general knowledge, neutral to negative perception and attitudes	moderate-to-high general knowledge positive perception and attitudes	Surveys providing qualitative and quantitative insight in change of attitudes and perception of targeted audiences	low general knowledge, neutral to negative perception and attitudes	low-to-moderate general knowledge, neutral perception and attitudes	moderate general knowledge, neutral to positive perception and attitudes	moderate general knowledge neutral to positive perception and attitudes	moderate-to-high general knowledge positive perception and attitudes	**LOW**• GED project missed to communicate to wider audience the benefits related to improving EE• The message is not communicated in a understandable manner	**LOW**• Due to UNDP experience on similar activities in BiH, it is not expected that the GED project will miss-out to communicate, create understanding and raise awareness on EE	• UNDP's communication & PR team supports the GED project throughout its raising awareness activities• All activities under the media campaign will be communicated in a understandable manner to the wider audience• Media buying and outreach are timely planned; with SWEDEN's involvement additional outreach and media campaign coverage will be assured
Output objective 5	**Public awareness on benefits of energy efficiency, energy management and reduction of emissions to air is increased.**											
	Indicator	**Baseline**	**Target**	**Source of verification**	**Results**	**Results**	**Results**	**Results**	**Results**			
		2014	**2018**		**2014**	**2015**	**2016**	**2017**	**2018**			
5.1.Indicator	Media campaign outreach achieved (seconds)	0	90000	Project reports and relevant technical documentation / Statistics of social networks access and use (shares, likes, comments)	18000	18000	18000	18000	18000			
5.2.Indicator	Number of awareness raising events held in BiH	0	75		15	15	15	15	15			
5.3.Indicator	Number of promotional materials distributed	0	150000		30000	30000	30000	30000	30000			
5.4.Indicator	Number of people reached through marketing campaign	0	500000		100000	100000	100000	100000	100000			
5.5.Indicator	Number of posts at GED public awareness campaign social networks (professor Atom)	0	4000		300	500	1000	1000	1500			